MIS Quarterly

RESEARCH ARTICLE

PRICE EFFECTS IN ONLINE PRODUCT REVIEWS: AN ANALYTICAL MODEL AND EMPIRICAL ANALYSIS[1]

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Abstract

Consumer reviews may reflect not only perceived quality but also the difference between quality and price (perceived value). In markets where product prices change frequently, these price-influenced reviews may be biased as a signal of product quality when used by consumers possessing no knowledge of historical prices. In this paper, we develop an analytical model that examines the impact of price-influenced reviews on firm optimal pricing and consumer welfare. We quantify the price effects in consumer reviews for different formats of review systems using actual market prices and online consumer ratings data collected for the digital camera market. Our empirical results suggest that unidimensional ratings, commonly used in most review systems, can be substantially biased by price effects. In fact, unidimensional ratings are more closely correlated with ratings of product value than ratings of product quality. Our findings suggest the importance for firms to account for these price effects in their overall marketing strategy and suggest that review systems could better serve consumers by explicitly expanding review dimensions to separate perceived value and perceived quality.

Keywords: Online product reviews, review bias, price effects, empirical analysis, optimal pricing

Introduction

In recent years, there has been growing research interest in examining dissemination of product information through online word-of-mouth. Consumers share product evaluations of a wide assortment of products through product review websites, discussion forums, blogs, and virtual communities. These networks serve many of the same functions as traditional word-of-mouth communications (Godes et al. 2005) that previously occurred only among family or friends. The large-scale experience-sharing among consumers in these networks potentially reduces uncertainty about the quality of products or services that cannot be inspected before purchase and therefore can play a substantial role in consumers' purchase decision processes. According to a survey by Deloitte's Consumer Products group (Deloitte 2007), almost two-thirds of consumers read consumer-written product reviews on the Internet. Among those consumers who read reviews, 82 percent say their purchase decisions have been directly influ-

[1]Chris Kemerer was the accepting senior editor for this paper. Mike Smith served as the associate editor.

The appendices for this paper are located in the "Online Supplements" section of the *MIS Quarterly*'s website (http://www.misq.org).

enced by the reviews and 69 percent share the reviews with friends, family, or colleagues, thus amplifying their impact. Other consulting reports and surveys have also shown that for some consumers and products, consumer-generated reviews are more valuable than expert reviews (ComScore 2007; Piller 1999), have a greater influence on purchasing decisions than traditional media (DoubleClick 2004), and have a significant impact on offline purchase behavior (ComScore 2007).

Information is being exchanged online in unprecedented scale and detail. This increased transparency makes it possible for researchers to monitor word-of-mouth over time and, therefore, to obtain a deeper understanding of consumer preferences and decision processes. The predominant research focus has been on the correlation between consumer reviews and sales (Chen et al. 2007; Chevalier and Mayzlin 2006; Clemons et al. 2006; Dellarocas et al. 2004; Duan et al. 2008; Forman et al. 2008; Godes and Mayzlin 2004). How well these review forums communicate product information, however, has been less studied. Li and Hitt (2008) found that because early adopters of products have different preferences for the underlying products, the reviews provided by these early adopters are not necessarily representative of the market as a whole. In addition, consumers do not appear to account for this review bias when they utilize these reviews to assist their purchase decisions. These findings suggest the importance of understanding the process by which consumers utilize review information, the process by which consumers decide to produce reviews, and the information that consumers convey through reviews. Although the former two issues are probably more amenable to laboratory study, the information content of reviews can be examined through a natural experiment present on the Internet. In particular, we examine in this study whether product price influences consumer reviews, and how firms can optimize their pricing strategy to take account of price effects in consumer reviews.

There are two reasons why price may play a role in consumer reviews. First, consumer reviews may reflect not only the perceived quality of a product or service but also the perceived value—the difference between the utility derived from product quality and price—from the purchase. Price has been shown to be a major influence on customer satisfaction in the manufacturing industry as a whole (Tsai 2007), as well as in service industries such as rental cars (McGregor et al. 2007). There is also anecdotal evidence that consumers take price into account when they write reviews. For example, in a review written on December 28, 2005, for the SONY Cyber Shot DSC-S40 digital camera posted on CNet.com, a consumer writes "some problems but at this price can't complain… .But for a 4 Mp camera at this price it is fantastic!" and gives a rating of 8 (out of 10) while for the same camera the CNet editor gives a rating of 6.6 (out of 10). Similar observations can be found in reviews for other products.

Second, previous research suggests that, when faced with quality uncertainty, consumers are likely to use price as a signal of quality (Dodds et al. 1991; Grewal 1995; Kirmani and Rao 2000; Mitra 1995; Olson 1997; Rao and Monroe 1988, 1989). Some of this quality expectation may be disconfirmed by actual experience. Because disconfirmation of prior expectations is known to influence satisfaction (Cadotte et al. 1987; Churchill and Surprenant 1982; Spreng et al. 1996; Rust et al. 1999), price may have an indirect effect on perceived quality, which is ultimately reflected in consumer reviews.

In this study, we first develop an analytical model of optimal firm pricing that accounts for the possibility that prices have an indirect effect on demand by altering consumer reviews. Next, to validate the assumptions of our model, we use actual market prices data and online consumer ratings data to empirically examine whether and to what extent price affects consumer ratings. Finally, we further validate our theoretical model predictions by examining whether the price–quality relationship anticipated in our theoretical model appears in the data.

Our empirical analysis on how prices affect reviews is done by comparing different review systems that cover the same product. Different review systems have different methods for collecting and displaying consumer review information. Most of these systems are one-dimensional, having only a single overall rating (e.g., CNet.com or Amazon.com for most products). Other systems divide ratings into multiple categories. For example, Dpreview.com, an online consumer review service focusing on digital cameras, divides consumer reviews into five categories: construction, features, image quality, ease of use, and value for money. Among them, the value for money dimension directly captures the role of price in ratings. For review systems with one-dimensional ratings, there is no explicit value for money component, but the influence of price may be directly incorporated into the single rating. By examining the relationship between reviews and prices over time and across different review systems, we can investigate the role of price in consumer valuation and product ratings. In particular, in this paper, we empirically examine the following assumptions and predictions of our analytical model:

(1) Does price affect consumer reviews?
(2) How is the role of price in consumer reviews different across different review systems?
(3) Are observed prices consistent with our model of how firms should respond to price effects in reviews?

Our empirical and analytical findings apply especially to markets where the product price changes frequently and at least some consumers are unable or unwilling (perhaps due to high

search costs) to seek historical prices to make proper adjustments. Understanding the role of price in affecting consumer reviews is potentially useful for websites designing review services to improve the efficiency of reviews in signaling product quality, and it is useful for firms attempting to understand the feedback provided by the market from product review sites to develop optimal pricing. For instance, our results suggest that firms can boost ratings of their products at release via low introductory pricing. Compared to other strategies for managing product reviews, such as hiring paid reviewers to create artificially high ratings (Dellarocas 2006; Mayzlin 2006), pricing strategically to influence consumer satisfaction at early stages of a product life cycle may be more cost-effective and less subject to ethical concerns.

Literature Review

Since Rogers (1962), word-of-mouth has been perceived as an important driver of sales in the product diffusion literature. Those models normally assume that consumers' experience with a product is communicated positively through word-of-mouth (Mahajan and Muller 1979) and therefore facilitates product diffusion. Prior empirical work on the relationship between word-of-mouth and product adoption usually measures the presence of word-of-mouth by inferring its existence and impact based on the opportunity for social contagion instead of observing it directly (Godes et al. 2005). For example, Reingen et al. (1984) infer word-of-mouth interaction based on whether individuals live together, and Foster and Rosenzweig (1995) infer knowledge spillover through word-of-mouth based on whether farmers live in the same village.

The emergence of large-scale online communication networks provides a channel for researchers to directly observe word-of-mouth over time and therefore to obtain a deeper understanding of consumer preferences and decision processes. Based on product reviews or conversation data collected from consumer networks, researchers are able to directly test the relationship between word-of-mouth and product sales in different industries. In the book industry, Chevalier and Mayzlin (2006) demonstrated that the differences between consumer reviews posted on the Barnes & Noble site and those posted on Amazon.com were positively related to the differences in book sales on the two websites. Two recent studies further found that reviews written by consumers from the same geographic location (Forman et al. 2008) or with a higher helpfulness vote (Chen et al. 2007) have a higher impact on sales. In the motion picture and television industries, Godes and Mayzlin (2004) showed a strong relationship between the popularity of a television show and the "dispersion" of conversations about TV shows across online consumer communities. Dellarocas et al. (2004) incorporated the sentiment of word-of-mouth into a product diffusion model and found that the valence (average numerical rating) of online consumer reviews is a better predictor of future movie revenues than other measures they considered. In contrast, Duan et al. (2008) suggested that the alternative causal relationship is also true: that the number of online reviews influences box office sales. In the beer industry, Clemons et al. (2006) found that the variance of ratings and the strength of the most positive quartile of reviews have a significant impact on the growth of craft beers.

Although the link between word-of-mouth and product sales was generally supported in the aforementioned studies, other researchers began to further examine this relationship by examining whether online reviews are effective in communicating actual product quality. Anderson (1998) proposed that consumers are more likely to engage in word-of-mouth when they have extreme opinions. Bowman and Narayandas (2001) further suggested that word-of-mouth behavior is also driven by customer loyalty to the brand. Li and Hitt (2008) found that consumers' self-selection behavior in purchasing may introduce bias into consumer reviews, which further affects sales and consumer welfare.

Built on the relationship between word-of-mouth and product sales, analytical studies of online reviews further examined how firms' profitability and marketing strategies can be affected. Dellarocas (2006) and Mayzlin (2006) examined the incentive of firms to manipulate reviews and the implications of this to consumer welfare. Chen and Xie (2008) showed that firms only have incentive to help disseminate consumer reviews when the firm's target market is sufficiently large, and that reviews affect the optimal product assortment and information provision policies of firms. Li et al. (2010) found that an S-shaped relationship exists between the quality of reviews and firm profits.

In all these existing studies, it is implicitly assumed that consumer reviews reflect consumers' perceptions of product quality and hence consumers are not affected by price. Whether this assumption is consistent with consumer behavior is left untested.

Zeithaml (1988, p. 14) defines value as "the consumer's overall assessment of the utility of a product based on perceptions of what is received and what is given." Value implicitly refers to a buyer's trade-off between benefit (quality) and cost (price) of the purchase (Bolton and Drew 1991). Although it is generally agreed that a consumer's purchase decision is determined by expected utility before purchase, perceived value from a purchase can also affect the consumer's *ex post* satisfaction with the purchase (Spreng et al. 1996). Correspondingly, price, in addition to quality, should also be an

important factor influencing consumers' post purchase satisfaction (Varki and Colgate 2001; Voss et al. 1998). Richins (1983) suggested that word-of-mouth may be driven by consumer satisfaction with a purchase. Therefore price, in turn, should influence consumer reviews.

A distinct line of argument suggests that consumers may use price as a signal of quality before they make purchase decisions when facing quality uncertainty (Dodds et al. 1991; Grewal 1995; Kirmani and Rao 2000; Mitra 1995; Olson 1997; Rao and Monroe 1988, 1989). Many studies have shown that consumers' post purchase satisfaction can be affected by the confirmation or disconfirmation of received quality after consuming the product versus their expectation before purchase (Cadotte et al. 1987; Churchill and Surprenant 1982; Rust et al. 1999; Spreng et al. 1996). Accordingly, by influencing consumers' expectations over quality before purchase, price may indirectly affect post-purchase consumer satisfaction and further affect consumer ratings.

In the next section, we first construct an analytical model that examines how firms can optimally price a product when price can have an effect on consumer reviews. In a later section, we empirically test our model assumptions and predictions using actual market prices data and online consumer ratings data collected for the digital camera market.

Analytical Model

In this section, we first examine firms' optimal pricing strategy in a monopoly setting and discuss the implications of the price effects in consumer reviews for consumer surplus. We then further verify the generalizability of our model predictions to duopoly settings.

Model Setup

We consider a two-period market for an experience good in which, in each period, a group of consumers comes into the market and makes a decision about whether to purchase up to one unit of the durable good with no repeat purchase opportunity. As suggested by standard product diffusion modeling approaches (e.g., Bass 1969), early adopters rely primarily on their own expectations, whereas later adopters can also be influenced by peer opinions such as consumer reviews.

The net utility of consuming the product for consumer i is defined as $U(x_i, q, p) = q - p - t\,x_i$, where p is the price of the product, which may vary over time. The value of element q measures the objective "quality" of the product and is the same for all consumers. To capture uncertainty in the quality of a product prior to purchase, q is a random draw from the interval [0, 1]. Consumers learn the actual value of q only after buying the product. To allow for horizontal differentiation in preferences for observable product attributes (e.g., color), we introduce a taste parameter (x_i), which represents the position of a consumer's ideal product in a product space (a bounded interval [0, 1]). The actual product occupies position 0 and this position is assumed exogenous (results are similar if we assume the actual product occupies position ½ instead). Consumers know the value of x_i before purchase and reduce their utility by a factor t per unit distance from the product, analogous to the travel distance in the Hotelling model (Hotelling 1929). We assume $t > 1$ so that, when combined with the fact that $q \leq 1$, a monopoly cannot cover the whole market.

In the first period, early adopters arrive and make their purchase decisions based on their expectation over quality (q^e) and first-period price (p_1). q^e is exogenously given and common across all consumers.[2] Without loss of generality, we normalize the value of the best alternative to this product to be zero. Thus, only consumers with expected utility $U(x_i, q^e, p_1)$ larger than zero will buy the product. The first period demand equals $(q^e - p_1)/t$. After consumers experience the product, they post their evaluations (R) online; the evaluations can be accessed by consumers who arrive in the second period. The second-period consumers will always wait for reviews to be posted before making purchase decisions. They update their quality perceptions based on these reviews, which affect their expected utility and therefore affect the second-period demand.

The distinctive feature of our model is that we allow reviews (i.e., reviewers' post purchase evaluations of the product) to be affected by both product quality as well as by the prices that reviewers paid. This assumption will be empirically tested using actual prices and ratings data (see the "Empirical Analysis" section). We normalize our rating measure such that it is numerically comparable to quality (bounded in the range [0, 1]), with R being equal to $\text{Max}\{0, \text{Min}\{1, q - b(p_1 - r(q))\}\}$. The parameter b captures the strength of the price effects ($0 < b < 1$) and $r(q)$ captures the market price for a product of quality q that is perceived as "reasonable" by consumers in the sense that deviations from this price have addi-

[2]Exogenous prior expectation assumption has appeared in prior studies on pricing of experience goods (e.g., Schmalensee 1982; Shapiro 1983a, 1983b; Villas-Boas 2004). Shapiro (1983b) points out that consumers' expectations about a new product's quality are generally not fully rational. Many factors can affect a consumer's expectation over a product's quality before purchase, such as advertising, prior experience with the brand, and average quality level of similar products in the market. If q^e differs across consumers, then we can include q_i^e in x_i, and the subsequent analysis still applies.

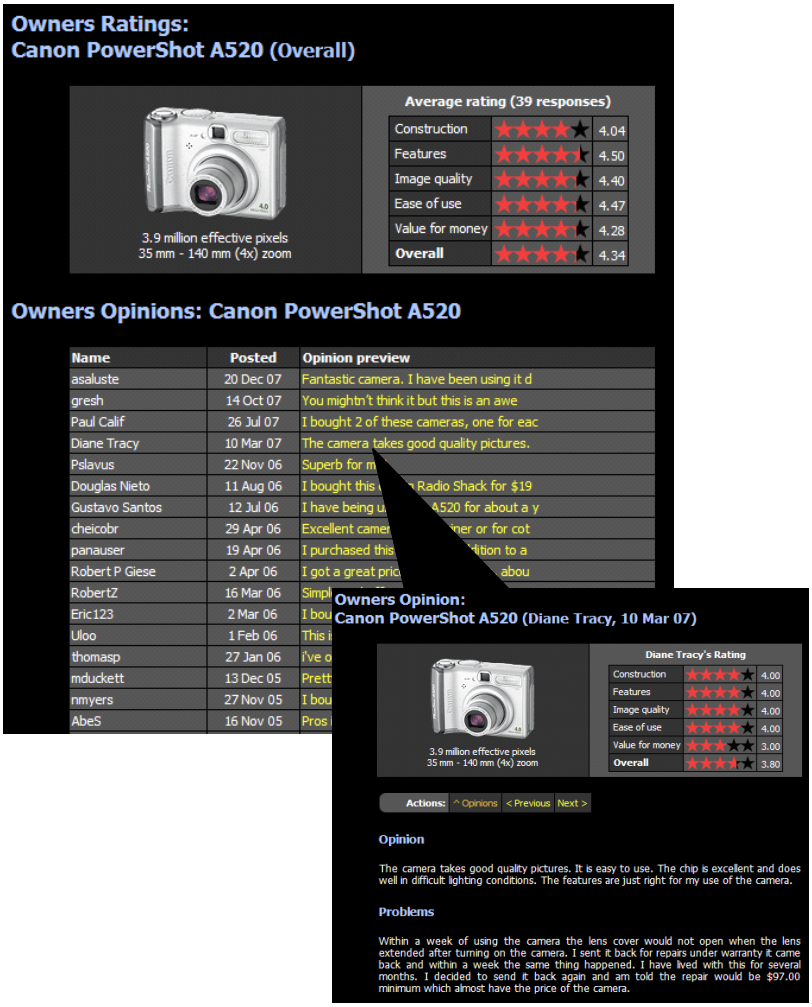


Figure 1. Sample Review Pages from Dpreview.com

tional effects on value perception. Thus, products with excessively high prices have their ratings (R) reduced by b per unit of price above the nominal level $r(q)$, while products that are less expensive receive a boost in their ratings of $b\,(r(q) - p_1)$. To provide some structure to $r(q)$, we assume $r(q)$ equals the standard monopoly price for products with quality q: $r(q) = q/2$. Our results do not appear to be sensitive to this assumption as long as $r(q)$ is an increasing function of q.[3]

In the second period, the "followers" arrive in the market and use reviews posted on different websites to form expectations of product quality before purchase. Depending on the type of websites they visit and whether they are willing to spend effort to seek historical price information, their expectations of quality may be influenced by reviews differently.

For our purposes, it is useful to divide consumer review systems into two types: those that provide only a single rating, and those that provide multiple rating dimensions. We are interested in comparing single rating systems to multidimensional rating systems that include a direct measure of consumers' value perceptions (the total utility of quality less price). Figure 1 shows a rating page from Dpreview.com, which has multidimensional ratings; Figure 2 shows a rating page from CNet.com, which utilizes a single review dimension.

[3]We also tried the assumption of $r(q)$ being a nonlinear function of q, which also turns R into a nonlinear function of q, and obtained very similar results. Derivation is available upon request from the authors.

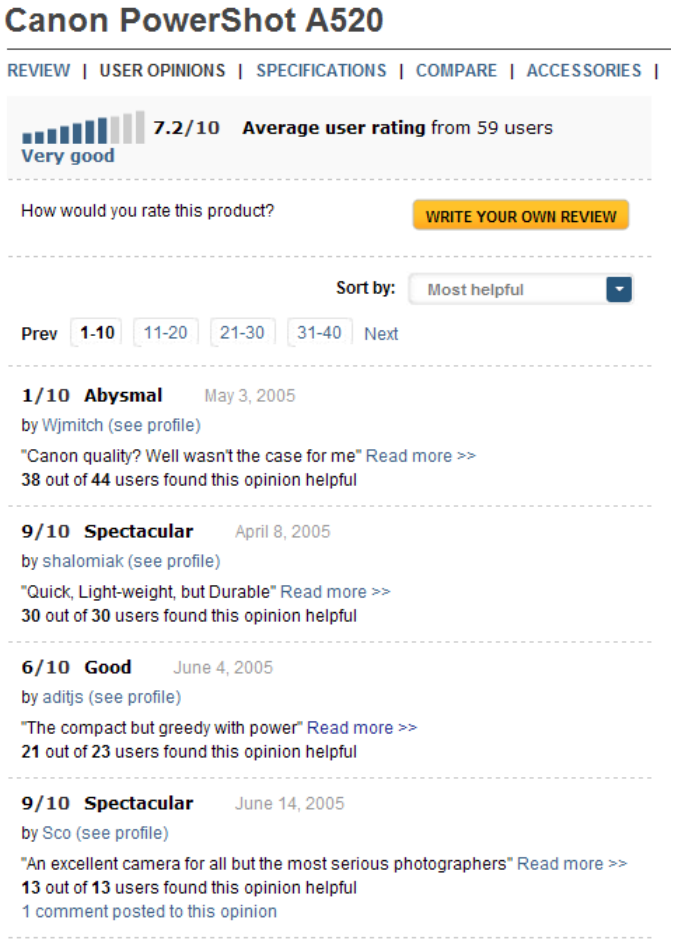


Figure 2. Sample User Opinions Page from CNet.com

Consumers who read reviews from websites that separate quality ratings from value ratings can derive q directly from average rating R even without knowledge of historical price. However, consumers who read reviews from websites that integrate product quality and influence of price into unidimensional ratings may not be aware of or are not able to derive q from R unless they are willing to incur a cost to retrieve historical price information.[4] The net effect of these behaviors, similar in spirit to search cost models, is that some fraction of consumers a ($0 < a < 1$) will be partially uninformed in the sense that they set their quality estimate to a value implied by the reviews they observe (R), while the remainder of consumers are fully informed and know q.[5]

[4]According to web browsing behavior tracked by ComScore in January 2006, more than 10 times as many camera buyers visited review websites with unidimensional ratings as those who visited review sites with multidimensional ratings (of the rating sites we consider), and 2.5 times as many camera buyers purchased from retailers who provide unidimensional ratings. Accordingly, if unidimensional ratings are indeed affected by price and are a biased signal of quality as demonstrated in our empirical analysis, this bias can have a significant impact on consumer purchasing behavior and thus on consumer welfare.

[5]For simplicity, we assume consumers' updated expectations are R (or q) without modeling the detailed belief updating process. If, alternatively, we assume that a consumer's prior expectation has a mean q^e and a variance σ_e^2, each review has a mean R (or q) and a variance σ_r^2, and the number of reviews is m, then following Bayes' Rule, the updated expectation will be

$$\frac{Rm^2\sigma_e^2 + q^e\sigma_\gamma^2}{m^2\sigma_e^2 + \sigma_\gamma^2} \text{ (or } \frac{qm^2\sigma_e^2 + q^e\sigma_\gamma^2}{m^2\sigma_e^2 + \sigma_\gamma^2}\text{)}$$

(DeGroot 1970, p. 168), which is a linear function of R (or q) and converges to R (or q) as m increases. Therefore, this simplified assumption will not affect our results qualitatively.

Many factors relevant to online consumer reviews may affect the size of our model parameter a. For example, access to expert ratings which focus primarily on quality (as will be shown in our empirical analysis for digital cameras) may enable more consumers to estimate q from unidimensional ratings, thus decreasing a. "Helpful" votes that are supported by many review websites to allow consumers to judge on the informativeness of the reviews (e.g., "38 out of 44 users found this opinion helpful" is listed along with the first review in Figure 2) may also affect parameter a to the extent that helpfulness or informativeness is evaluated based on the ability of the reviews to help users of the reviews to separate quality information from price information. Thus, parameter a can be considered as the result of all of these different mechanisms involved in the consumer's perception formation process.

All consumers will make their purchase decisions based on the expected product quality and the second-period price p_2. Let n represent the ratio of the number of consumers in the market in the second period to the number of consumers in the market in the first period (to allow for market growth or shrinkage). The firm selects p_1 and p_2 to maximize its total profit:

$$\frac{p_1(q^e - p_1)}{t} + n\left((1-a)p_2 Max\left\{0, \frac{q-p_2}{t}\right\} + ap_2 Max\left\{0, \frac{R-p_2}{t}\right\}\right) \quad (1)$$

where $\begin{cases} R = Max\{0, Min\{1, q - b(p_1 - r(q))\}\} \\ r(q) = \frac{q}{2} \\ p_1 < q^e \\ p_2 < Max\{q, R\} \end{cases}$

The optimal prices (p_1^* and p_2^*) and average rating under optimal prices (R^*) are all functions of quality q and parameters a, b, q^e, and n:

$$p_1^* = \begin{cases} \frac{q^e}{2} & \text{if } 0 < q \le \bar{Q}_1 \text{ or } \frac{2+bq^e}{2+b} \le q < 1 \\ \text{Max}\left\{0, \frac{(2+b)q-2}{2b}\right\} & \text{if Max}\left\{\bar{Q}_1, \text{Min}\left\{\bar{Q}_2, \frac{4q^e}{2ab n + a^2b^2 n}\right\}\right\} \le q < \frac{2+bq^e}{2+b} \\ \frac{q^e}{2} - \frac{ab((2+ab)q - abq^e)n}{2(4-a^2b^2n)} & \text{if } \bar{Q}_1 < q < \text{Max}\left\{\bar{Q}_1, \text{Min}\left\{\bar{Q}_2, \frac{4q^e}{2ab n + a^2b^2 n}\right\}\right\} \end{cases}$$

$$p_2^* = \begin{cases} \frac{a+(1-a)q}{2} & \text{if Max}\left\{\bar{Q}_2, \frac{2}{2+b}\right\} < q < 1 \\ \frac{q}{2} + \frac{ab(q-2p_1^*)}{4} & \text{if } \bar{Q}_1 < q < \text{Max}\left\{\bar{Q}_2, \frac{2}{2+b}\right\} \\ \frac{q}{2} & \text{if } 0 < q < \bar{Q}_1 \end{cases}$$

$$R^* = \begin{cases} 1 & \text{if Max}\left\{\bar{Q}_2, \frac{2}{2+b}\right\} < q < 1 \\ q - b(p_1^* - \frac{q}{2}) & \text{if } \bar{Q}_1 < q < \text{Max}\left\{\bar{Q}_2, \frac{2}{2+b}\right\} \\ 0 & \text{if } 0 < q < \bar{Q}_1 \end{cases}$$

where

$$\bar{Q}_1 = \begin{cases} \frac{2q^e}{\sqrt{a(4+b(4+ab))n}} & if\ n > \frac{4(4+4b+ab^2)}{ab^2(2+ab)^2} \\ \frac{bq^e\left(2+ab+\sqrt{(1-a)(4-a^2b^2n)}\right)}{4+b(4+ab(1+(1-a)n))} & if\ n < \frac{4(4+4b+ab^2)}{ab^2(2+ab)^2} \end{cases},$$

$$\bar{Q}_2 = \frac{4 - a^2b^2n + 2bq^e}{4+b(2+(1-a)abn)}.$$

Please refer to Appendix A for the derivation of optimal price functions. In comparison, we also derive optimal prices for the benchmark scenario in which reviews reflect purely quality and are not affected by prices. In the benchmark scenario, the firm selects p_1 and p_2 to maximize its total profit of $\frac{p_1(q^e - p_1)}{t} + \frac{np_2(q - p_2)}{t}$ given $p_1 < q^e$ and $p_2 < q$. It can be easily shown that the optimal first period price is $q^e/2$, the second period price is $q/2$, and the profit is $\frac{q^{e2} + nq^2}{4t}$. For illustrative purposes, the optimal price functions and rating for our model (1) and their comparison with the benchmark scenario are graphically shown in Figure 3 for a given numerical example: $a = 0.8$, $b = 0.8$, $q^e = 0.5$ and $n = 3$. Trends are similar for other values of parameters (a, b, q^e, and n).

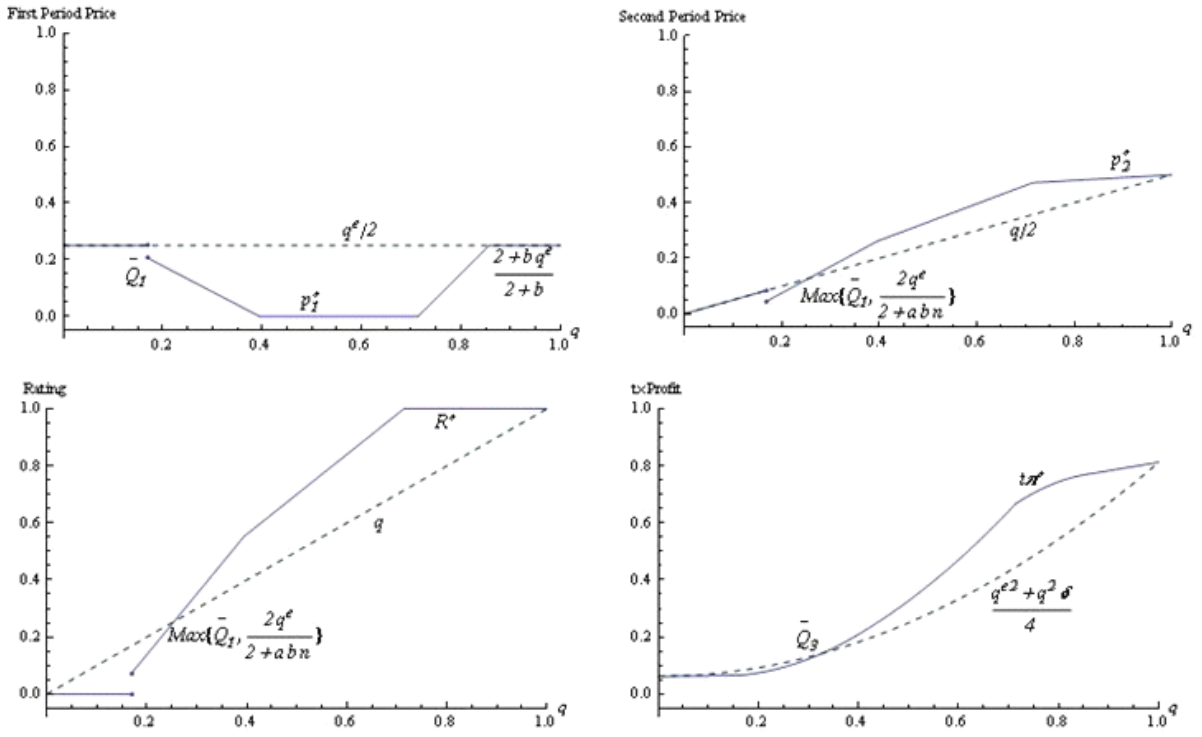
Model Predictions: Pricing, Profitability, and Rating

By comparing the optimal prices, profit and rating derived from model (1) to those in the benchmark scenario, we derive the following implications on the impact of price-influenced ratings (see Figure 3):

(1) Firms have the highest incentive to lower price in the first period to boost ratings in the second period if quality (q) is neither very high nor very low ($\bar{Q}_1 < q < \frac{2+bq^e}{2+b}$).

(2) Rating is higher than quality (q) only if quality is relatively high ($q > Max\left\{\bar{Q}_1, \frac{2q^e}{2+abn}\right\}$).

(3) Firms are able to charge a higher price in the second period when reviews reflect both price and quality than when reviews are pure quality measures only if quality (q) is relatively high ($q > Max\left\{\bar{Q}_1, \frac{2q^e}{2+abn}\right\}$).



Solid line: prices influence reviews
Dotted line: no price effects

$a = 0.8$, $b = 0.8$, $q^e = 0.5$, and $n = 3$

Figure 3. Comparison of Optimal Monopoly Prices, Rating, and Profit between Our Model (Prices Influence Reviews) and the Benchmark (No Price Effects)

(4) Only firms selling high-quality products produce higher profit ($q > \bar{Q}_3$) when reviews reflect both price and quality than when reviews are pure quality measures.[6]

The intuition behind this set of pricing results can be explained as follows. If product quality is very high ($q > \frac{2+bq^e}{2+b}$), the firm can achieve maximal ratings without having to discount the price at all. This is because quality uncertainty in the first period encourages the firm to price lower than would be optimal under full information, and this reduced price gives consumers the perception that the product was not only of high quality but also of high value given the price. This boost from the price component raises reviews over what they would be if only quality mattered, allowing the firm to increase price above the second period benchmark level (of no price influence in reviews).

Firms selling low-quality products face a strategic decision: whether to exploit quality uncertainty in the first period by charging high prices and then lose sales in the second period from the additional negative effects of high prices on reviews, or to price low to try to create high product reviews. When q is very low ($q < \bar{Q}_1$), the former incentive dominates. The firm garners monopoly revenue in the first period but loses in the second period because of downward-biased ratings influenced by price. As a result, the total profit is lower than that in the benchmark scenario.

If product quality is in the middle range ($\bar{Q}_1 < q < \frac{2+bq^e}{2+b}$), then the firm has an incentive to lower the price in the first period to boost ratings in the second period. Whether the firm

[6] $\bar{Q}_3 = \begin{cases} \dfrac{q^e\left(2+ab+\sqrt{4-a^2b^2n}\right)}{4+ab(1+n)} & \text{if } n < \dfrac{4(4+ab)}{ab(2+ab)^2} \\ \dfrac{2q^e}{\sqrt{(4+ab)abn}} & \text{if } n > \dfrac{4(4+ab)}{ab(2+ab)^2} \end{cases}$ and

$0 < Max\left\{\bar{Q}_1, \frac{2q^e}{2+abn}\right\} < \bar{Q}_3 < q^e < \frac{2+bq^e}{2+b} < 1.$

benefits from having price-influenced reviews versus quality-only reviews is indeterminate. If quality is on the higher end of this middle range ($q > \bar{Q}_3$), the second period profit gain offsets the loss from underpricing in the first period; otherwise, the firm is worse off. This is assuming optimal response. If the price influences reviews and firms ignore that fact, profits are further reduced in either setting for firms selling middle-quality products.

A consequence of the firm's optimal response to price-influenced reviews is that the average rating moves toward the extremes (see Figure 3): it is higher than quality q if q is high ($q > Max\left\{\bar{Q}_1, \frac{2q^e}{2+abn}\right\}$) and lower than quality q if q is low ($q < Max\left\{\bar{Q}_1, \frac{2q^e}{2+abn}\right\}$). The dividing threshold ($Max\left\{\bar{Q}_1, \frac{2q^e}{2+abn}\right\}$) is a function of a, b, q^e, and n. It increases with consumers' initial expectations (q^e) and decreases with the percentage of consumers (a) following price-influenced ratings, with the magnitude of price's impact on rating (b), and with the number of followers relative to the number of early adopters (n). Accordingly, upward bias in ratings is most pronounced when initial expectations are low, large numbers of consumers are partially informed (know R but not q), price has a large influence on ratings, and the market is growing more rapidly such that the second period market size is considerably larger than the first.

The firm's optimal response to price-influenced reviews also suggests the potential impact of these price-influenced ratings on consumer welfare, which can be inconclusive, contingent on the product quality q and the value of model parameters a, b, q^e, and n: consumers may benefit from reduced first-period prices or be hurt by biased reviews. These implications are further discussed in the next section.

Model Predictions: Consumer Welfare

In examining the welfare implications of this model, it is useful to distinguish the effects on early adopters (who might benefit from reduced first-period prices) from later adopters (for whom the firm no longer has a review-related incentive to underprice). For very high or very low quality, the first-period price is unaffected by the review regime (whether or not price affects ratings), so consumer surplus of first- period adopters is the same. If product quality is in the middle range discussed earlier ($\bar{Q}_1 < q < \frac{2+bq^e}{2+b}$), the firm has a benefit from lowering prices and therefore first-period consumers receive greater surplus. This region ($\bar{Q}_1 < q < \frac{2+bq^e}{2+b}$) is largest when the market is growing (n is large) and a large number of consumers are imperfectly informed (a is large).

Although early adopters may be weakly better off due to the firm's incentive to manipulate ratings by lowering early price, the followers who buy in the second period may become the victims of these price-influenced ratings. Figure 4 compares the total consumer surplus in both periods with the benchmark scenario (in which reviews are not affected by pricing) assuming that the firm follows optimal pricing in both scenarios. In all four figures, when parameters fall in the gray area, consumers are worse off compared to the benchmark scenario; when parameters fall in the white area, consumers are better off. The region of reduced consumer surplus (gray area) is larger when a, b, and n increase or q^e decreases. Accordingly, a larger number of consumers following price-influenced ratings, a greater impact of price on ratings, a larger number of consumers purchasing in the second period, and lower overall product quality (and thus lower prior expectation) tend to lead to lower consumer surplus.

It is worth noticing that the consumers who did not follow price-influenced ratings, either by reading reviews posted on websites that separate quality ratings from value ratings or by spending effort to seek historical price information, may also be negatively affected by the existence of these ratings because of the increased second-period price. That is, the existence of price-influenced ratings and the consumers who follow them may impose a negative externality on other consumers who also arrive in the second period. As we discussed earlier, early adopters may benefit from these price-influenced ratings and therefore would have little incentive to correct this bias when they write reviews, which may further aggravate the problem.

Model Extension: Duopoly Competition

In previous sections, we modeled the firm's optimal response to price-influenced reviews in a monopoly setting. In this section, we further extend our model to examine whether our model predictions derived in the monopoly setting can be generalized to a competitive market.

Consider two competing firms, Firm 1 and Firm 2, producing two experience goods at zero marginal cost. Each consumer is assumed to buy at most one unit of the product sold by either Firm 1 or Firm 2. Consistent with our assumptions in

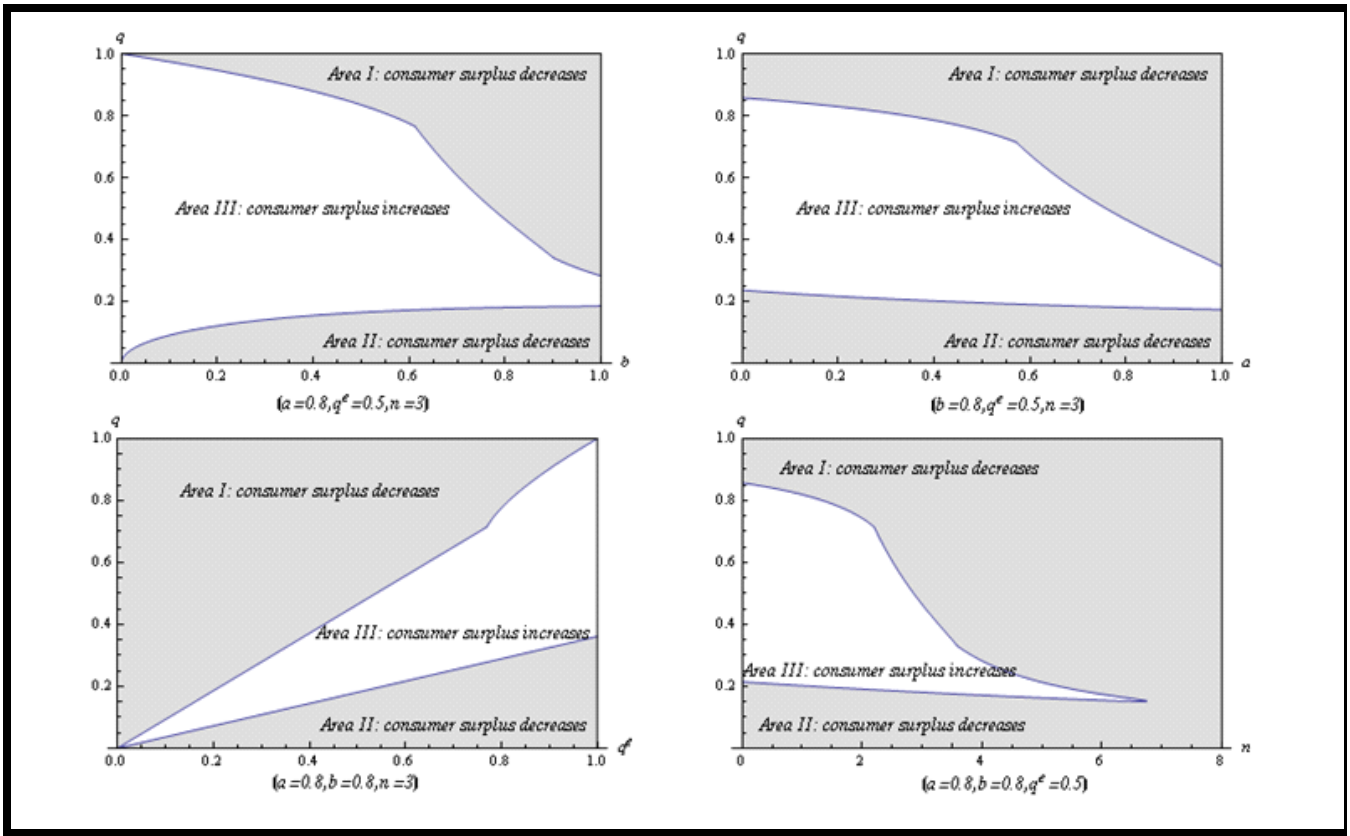


Figure 4. Comparison of Consumer Surplus in Our Model to that in the Benchmark Scenario (No Price Effects in Reviews)

previous sections, we assume that consumers are uniformly distributed along a unit line in preference space with taste parameter x_i representing the position of consumer i's ideal product. Firm 1's product is located at position 0 and Firm 2's product is located at position 1. For a buyer located at x_i on the unit line, her surplus from purchasing Firm 1's product is $q_1 - tx_i - p_1$, and her surplus from purchasing Firm 2's product is $q_2 - t(1 - x_i) - p_2$, where p_j (j = 1,2) is the price of Firm j, q_j ($q_j \in [0, 1]$) measures the objective "quality" of Firm j's product, and t is the disutility for a unit distance in the preference space incurred due to the mismatching between the position of the consumer's ideal product and the position of the firm's product in the preference space.

If t is sufficiently large, then the two products are sufficiently differentiated. In this case, both firms will behave as monopolists in both periods and our results from previous sections can apply directly. To check whether our results also apply to competitive markets where firms' potential markets overlap, in this section, we focus on the setting where t is sufficiently small such that all first period consumers buy from one of the firms. That is, neither firm is able to charge a monopoly price in the first period. Let p_{jk} (j = 1, 2; k = 1, 2) represent Firm j's price in period k, q_j^e represent consumers' expectation over the quality of Firm j's product before access to consumer reviews, and R_j represent the rating of Firm j's product posted by early adopters.

In the first period, early adopters arrive and purchase the product with higher positive expected utility based on their expectations over the quality of the two products (q_1^e and q_2^e) and the first period prices (p_{11} and p_{21}). After these early adopters consume the products, they post their evaluations online for the product they purchased (R_1 and R_2). Second-period consumers will always wait for reviews of both products to be posted before making purchase decisions. They update their quality expectations based on these reviews, which affect their expected utility and thus the second-period demand. Firms select their prices in both periods strategically to maximize their total intertemporal profits.

We solve the game for a simplified case ($a = 1$, $b = 1$, $n = 3$, $t = \frac{1}{6}$, $q_1^e = q_2^e = \frac{1}{2}$) to demonstrate that our results from previous sections can be generalized to sufficiently competitive markets. Similar to our results (1) – (4) (in the earlier section on pricing, profitability, and rating), we examine how a firm's optimal prices, rating, and profit are affected by its own product quality given all other parameters are fixed. To do this, without loss of generality, we fix Firm 1's quality to be 1, $\frac{1}{2}$, and 0, respectively, and examine how Firm 2's optimal prices, rating, and profit change as its quality varies from 0 to 1. Please refer to Appendix B for the derivation of the equilibria. Firm 2's equilibrium prices (p_{21}^* and p_{22}^*) and rating (R_2^*) are all functions of q_2, which vary with the value of q_1:

(1) If $q_1 = 1$

$$p_{21}^* = \begin{cases} \frac{1}{6} & \text{if } \frac{7}{9} < q_2 < 1 \\ \frac{3q_2-2}{2} & \text{if } \frac{2}{3} < q_2 < \frac{7}{9} \\ 0 & \text{if } \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{7-12q_2}{10} & \text{if } \frac{4}{9} < q_2 < \frac{7}{12} \\ \frac{1}{6} & \text{if } 0 < q_2 < \frac{4}{9} \end{cases}, p_{22}^* = \begin{cases} \frac{1}{6} & \text{if } \frac{2}{3} < q_2 < 1 \\ \frac{3q_2-1}{6} & \text{if } \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{9q_2-4}{10} & \text{if } \frac{4}{9} < q_2 < \frac{7}{12} \\ 0 & \text{if } 0 < q_2 < \frac{4}{9} \end{cases},$$

$$R_2^* = \begin{cases} 1 & \text{if } \frac{2}{3} < q_2 < 1 \\ \frac{3q_2}{2} & \text{if } \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{27q_2-7}{10} & \text{if } \frac{4}{9} < q_2 < \frac{7}{12} \\ \frac{3q_2}{2} - \frac{1}{6} & \text{if } \frac{1}{9} < q_2 < \frac{4}{9} \\ 0 & \text{if } 0 < q_2 < \frac{1}{9} \end{cases}$$

(2) If $q_1 = \frac{1}{2}$

$$p_{21}^* = \begin{cases} \frac{2}{15} & if\ \frac{34}{45} < q_2 < 1 \\ \frac{3q_2-2}{2} & if\ \frac{2}{3} < q_2 < \frac{34}{45} \\ 0 & if\ \frac{1}{3} < q_2 < \frac{2}{3} \\ \frac{-3q_2+1}{4} & if\ \frac{2}{9} < q_2 < \frac{1}{3} \\ \frac{1}{12} & if\ 0 < q_2 < \frac{2}{9} \end{cases}, p_{22}^* = \begin{cases} \frac{17}{60} & if\ \frac{34}{45} < q_2 < 1 \\ \frac{3q_2}{8} & if\ \frac{2}{3} < q_2 < \frac{34}{45} \\ \frac{6q_2-1}{12} & if\ \frac{1}{3} < q_2 < \frac{2}{3} \\ -\frac{1}{6} + \frac{3q_2}{4} & if\ \frac{2}{9} < q_2 < \frac{1}{3} \\ 0 & if\ 0 < q_2 < \frac{2}{9} \end{cases},$$

$$R_2^* = \begin{cases} 1 & if\ \frac{2}{3} < q_2 < 1 \\ \frac{3q_2}{2} & if\ \frac{1}{3} < q_2 < \frac{2}{3} \\ \frac{9q_2-1}{4} & if\ \frac{2}{9} < q_2 < \frac{1}{3} \\ \frac{3q_2}{2} - \frac{1}{12} & if\ \frac{1}{18} < q_2 < \frac{2}{9} \\ 0 & if\ 0 < q_2 < \frac{1}{18} \end{cases}$$

(3) If $q_1 = 0$

$$p_{21}^* = \begin{cases} \frac{1}{6} & if\ \frac{7}{9} < q_2 < 1 \\ \frac{3q_2-2}{2} & if\ \frac{2}{3} < q_2 < \frac{7}{9} \\ 0 & if\ \frac{2}{9\sqrt{6}} < q_2 < \frac{2}{3} \\ \frac{1}{6} & if\ 0 < q_2 < \frac{2}{9\sqrt{6}} \end{cases}, p_{22}^* = \begin{cases} \frac{5}{6} & if\ \frac{2}{3} < q_2 < 1 \\ \frac{3q_2}{2} - \frac{1}{6} & if\ \frac{2}{9} < q_2 < \frac{2}{3} \\ \frac{3q_2}{4} & if\ \frac{2}{9\sqrt{6}} < q_2 < \frac{2}{9} \\ 0 & if\ 0 < q_2 < \frac{2}{9\sqrt{6}} \end{cases},$$

$$R_2^* = \begin{cases} 1 & if\ \frac{2}{3} < q_2 < 1 \\ \frac{3q_2}{2} & if\ \frac{2}{9\sqrt{6}} < q_2 < \frac{2}{3} \\ 0 & if\ 0 < q_2 < \frac{2}{9\sqrt{6}} \end{cases}$$

In comparison, we also derive optimal prices for the corresponding benchmark scenarios in which reviews reflect pure quality and are not affected by prices. Results for these benchmark scenarios are also provided in Appendix B. For illustrative purposes, the optimal price functions and rating in this competitive setting and their comparison with the benchmark scenarios are graphically shown in Figure 5. These trends are very similar to those in Figure 3 for the monopoly setting. Our findings (1) – (4) (in the earlier section) all hold here (with different thresholds) except that in the competitive setting, when facing competitors with mid-quality products, firms with very high quality products actually earn less profit when reviews reflect both price and quality than when reviews are pure quality measures. This is because firms with mid-quality products have a higher incentive to under-price in the first period to boost ratings. When high quality firms face a mid-quality competitor, they have an incentive to lower the price to be competitive, but, unlike their mid-quality competitor, get very little ratings benefit from doing so since they are already close to maximum ratings.

Empirical Analysis

Our analytical model suggests that price-influenced ratings can affect optimal firm behavior and lead to reduced consumer surplus in many instances. In this section, we empirically test our model assumptions using actual ratings and prices data in the digital camera market.

Hypotheses

In our model, we assume that consumer reviews reflect not only reviewers' evaluations of product quality but also price. To empirically test this assumption, we can utilize the structure from our analytical model to model ratings as a function of product price and quality perceived by the reviewers:

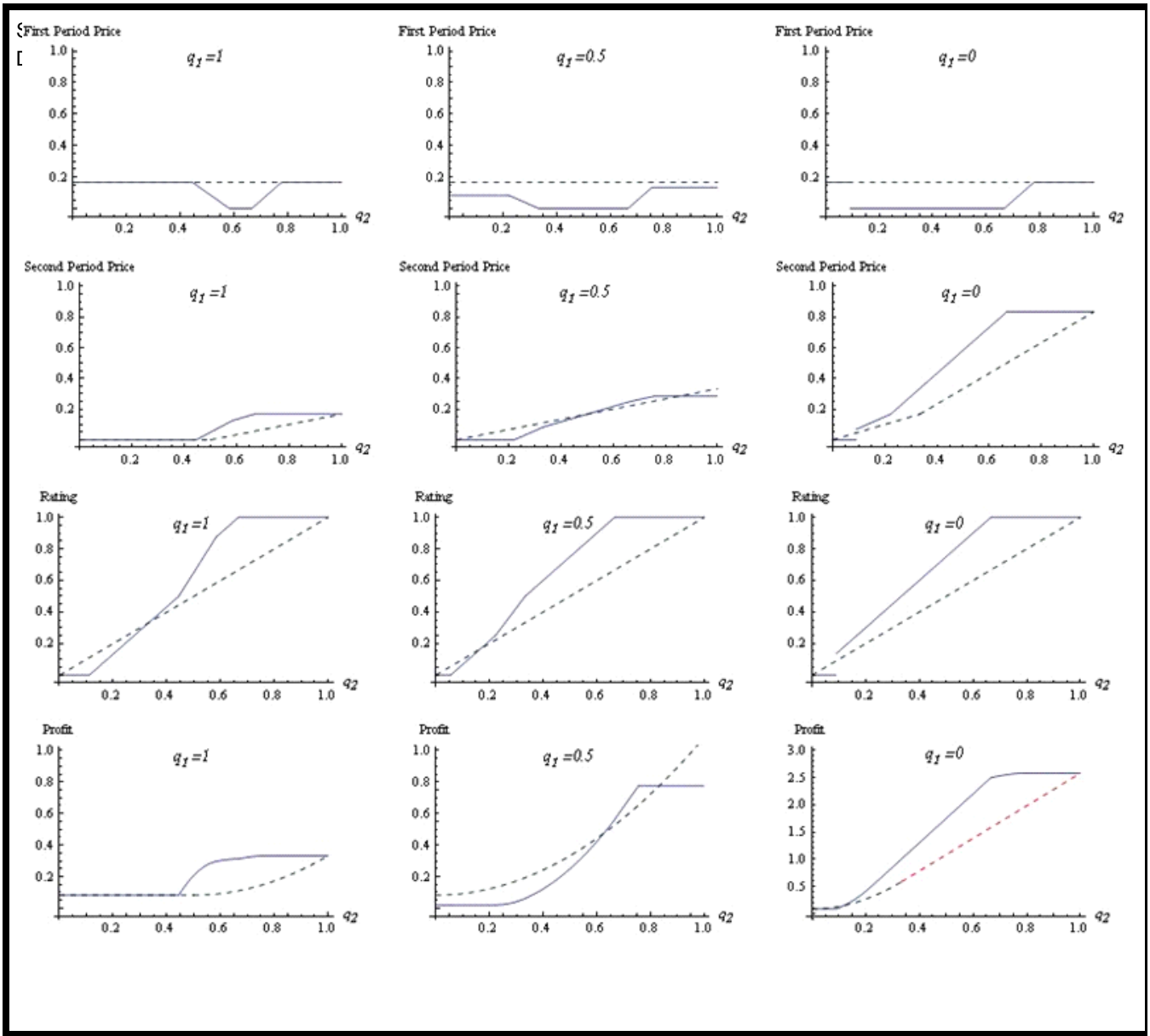


Figure 5. Comparison of Optimal Duopoly Prices, Rating, and Profit in Our Model to Those in the Benchmark Scenario

$$\text{Rating} = \beta_1 \text{ Quality} + \beta_2 \text{ Price}$$

Here, β_1 would correspond to $1 + \frac{b}{2}$, and β_2 would correspond to $-b$ in our model, suggesting that $\beta_1 > 0$ and $\beta_2 < 0$. Sites that have multidimensional reviews, such as Dpreview.com, have separate quality ratings and value ratings, whereas unidimensional review sites, such as CNet.com, integrate each reviewer's evaluation of all aspects into one single rating. If our assumption is valid, one should expect both unidimensional ratings and value ratings in multidimensional reviews to be negatively affected by price, but not quality ratings in multidimensional reviews. Thus, our first set of hypotheses related to our model assumption can be formulated as follows:

H1. Value ratings provided by multidimensional review sites are negatively affected by the price paid by the reviewers.

H2. Quality ratings provided by multidimensional review sites are not affected by the price paid by the reviewers.

H3. Ratings provided by unidimensional review sites are negatively affected by the price paid by the reviewers.

To validate our model assumption, we can also empirically test whether observed market prices are consistent with our earlier model predictions on pricing. This test is somewhat more challenging because our two-period model is only an approximation of the actual structure of the market, and product heterogeneity may mask some of the pricing behaviors we seek to observe. Nonetheless, our model does have a strong prediction that if firms take into account the impact of price-influenced ratings on demand when setting their prices, then firms in the middle quality range have the strongest incentives to price strategically (that is, to underprice in earlier periods and raise prices later to harvest the benefits of higher reviews), while this incentive is not present for firms with very high and very low quality products. Tests of this prediction are confounded, however, given the lack of clear distinctions between the first period and second period. Nonetheless, if this strategic behavior is being utilized in practice, it should have two observable characteristics. First, we should expect middle-quality firms on average to price lower for a given quality than high- or low-quality firms (giving rise to an upward U-shaped relationship between price and quality).[7] Second, we expect middle-quality firms to have a greater variation in price over time (an inverted U-shaped relationship between price variance and quality). These two predictions on the shape of the price–quality relationship are less sensitive to model assumptions than the results on exact prices, and therefore will be tested as a further validation of our model.

H4. There exists an upward U-shaped relationship between product price and product quality.

H5. There exists a downward U-shaped relationship between variance in product price and product quality.

[7]Without a clear distinction between the first and second period, this U-shaped relationship will appear flatter in the data than the actual price strategy. For firms selling middle-quality products, some will be in the second period while others will be in the price-lowering first period. The average for all firms will lie in the interval. Thus, our estimates of the strength of this relationship will be biased downward, making them more conservative. A significant curvature shows a potentially strong correspondence with our theoretical prediction.

Data Description

Our data consist of 88 digital cameras[8] released between 2002 and 2005 by Canon, Casio, Fujifilm, HP, Kodak, Konica, Kyocera, Nikon, Olympus, Panasonic, Pentax, and Sony. For each camera, the data consist of monthly unit sales, dollar sales, and average transaction prices on the market level collected by a market research company, NPD Group, from January 2002 to August 2005. For each camera, we collected camera features, including brand name, model name, release date, resolution, format, maximum shutter speed, LCD size, LCD resolution, digital zoom, optical zoom,[9] storage included, and whether battery charger, manual focus, and image stabilization functions are included. Using software agents, we collected all consumer reviews posted on CNet.com and Dpreview.com from the time the camera was released until August 2005. Figures 1 and 2 present examples of reviews posted on CNet.com and Dpreview.com, respectively. Reviews posted on CNet.com have single-value ratings (ratings range from 1 to 10), whereas reviews posted on Dpreview.com subdivide ratings into six categories: construction, features, image quality, ease of use, value for money, and overall (overall is the average of the first five). We suspect that the first four categories mostly reflect product quality, while the value-for-money category reflects perceived value from the purchase—that is, the tradeoff between quality and price. Ratings for each of the five categories range from 1 to 5. Among the 88 cameras, 80 cameras also received editor ratings assigned by CNet editors before August 2005, which were also collected. These two websites were selected for this study because of their popularity: they are among the top three ranked websites shown in the main search results on Google.com when searching for "digital camera reviews" or "digital cameras reviews" (based on an informal experiment run by the authors).

[8]Our data initially had 490 cameras (including all the cameras that were released between 2002 and 2005 and had reviews posted on Dpreview.com before August 2005), but not all cameras have ratings from CNet.com nor have corresponding market prices data. In this study, because we need to compare reviews across websites and also examine the relationship between ratings and prices, our final dataset contains only the 88 cameras that have ratings from both websites as well as market prices data. Although not all the cameras that are offered are included, a preliminary sample-selection analysis suggests that our data set is representative of the cameras with different market sizes. The only segment not represented by our sample is the group of cameras with less than 2,000 units sold during our data collection periods.

[9]Digital zoom and optical zoom are available only for non-SLR cameras. For SLR cameras, digital zoom and optical zoom vary with the lens, which consumers buy separately.

Table 1. Descriptive Statistics for Monthly Ratings and Price Data

Variables		Number of Observations	Mean	Standard Deviation	Minimum	25% Percentile	Median	75% Percentile	Maximum
Dpreview (per-month cumulative average)	Average Rating for Construction	862	4.42	0.29	2.93	4.27	4.44	4.60	5
	Average Rating for Features	862	4.38	0.33	2.64	4.27	4.44	4.61	5
	Average Rating for Image Quality	862	4.32	0.39	2.7	4.17	4.40	4.58	5
	Average Rating for Ease of Use	862	4.41	0.29	3	4.26	4.46	4.58	5
	Average Rating for Value for Money	862	4.33	0.34	2.57	4.19	4.39	4.56	5
	Average Rating for Overall	862	4.37	0.27	2.91	4.28	4.43	4.55	4.9
	Number of Ratings	862	45	53	1	11	26	60	332
CNet (per-month cumulative average)	Average Rating	897	7.85	0.70	5	7.5	7.92	8.27	10
	Number of Ratings	897	36	38	1	12	24	47	214
Average Market Price		1238	$435	$278	$70	272	356	491	$1696

These two websites provided a total of 8,320 single review observations. We then aggregate these data to represent a per-month (cumulative) average for all of our calculations, which yields a total of 897 observations of average ratings for CNet and 862 observations of average ratings for Dpreview. The number of observations differs between the two websites because during certain months, reviews may be posted to one website but not the other. Table 1 provides the summary statistics for this sample.

Decomposing Multidimensional Ratings into Price and Quality Components

To test hypotheses H1, first of all, we need to control for other factors that may affect value ratings, among which the most important is consumer's perception of product quality. To capture product quality, we utilize the four dimensions other than value for money from Dpreview (construction, features, image quality, and ease of use). This assumption will also be empirically tested later in this section. To control for heterogeneity among cameras, we also include covariates for common objectively observable camera features (format: ultra compact, compact, or SLR; maximum shutter speed, LCD size, LCD resolution, digital zoom, optical zoom, storage included; the inclusion of battery charger; the availability of manual focus; the availability of image stabilization; and image resolution) as well as month and brand dummy variables. These controls remove variation in price and quality due to objective differences among products, thus enabling the marginal effect of price on review to be more comparable across products. This yields the following estimating equation:[10]

$$\begin{aligned}\text{ValueForMoney}_{it} &= \beta_0 + \beta_{11}\text{Construction}_{it} + \beta_{12}\text{Features}_{it} \\ &+ \beta_{13}\text{ImageQuality}_{it} + \beta_{14}\text{EaseOfUse}_{it} + \beta_2\text{Log}(\text{AvgPrice}_{it-1}) \\ &+ \alpha_1\text{FormatDum1}_i + \alpha_2\text{FormatDum2}_i + \alpha_3\text{Log}(1/\text{MaxShutter}_i) \\ &+ \alpha_4\text{LCDSize}_i + \alpha_5\text{LCDResolution}_i + \alpha_6\text{DigitalZoom}_i \\ &+ \alpha_7\text{OpticalZoom}_i + \alpha_8\ \text{StorageIncluded}_i \\ &+ \alpha_9\text{BatteryChargerDummy}_i + \alpha_{10}\ \text{ManualFocusDummy}_i \\ &+ \alpha_{11}\text{ImageStabilizationDummy}_i + \lambda_1\text{ResolutionDummies}_i \\ &+ \lambda_2\text{BrandDummies}_i + \lambda_3\text{MonthDummies}_t + \varepsilon_{it}\end{aligned} \quad (2)$$

The estimated coefficient β_2 indicates the impact of price on consumer's value function after controlling for objective and subjective measures of quality and observable camera features.

[10]Logarithmic transformation is applied on 1/Max Shutter to achieve normal distribution (Gelman and Hill 2007).

For empirical implementation, each review component variable ($ValueforMoney_{it}$, $Construction_{it}$, $Features_{it}$, $ImageQuality_{it}$, and $EaseOfUse_{it}$) represents the cumulative average rating of camera i from release until month t. Because we are unable to observe the price paid by each reviewer, we instead construct a measure of the average price paid for the camera from release until the prior month ($t-1$) to obtain $AvgPrice_{it-1}$. By applying a logarithmic transformation, the coefficient on price represents the change in review per percentage change in price:

$$\frac{\partial \text{ValueForMoney}_{it}}{\partial \text{AvgPrice}_{it-1}} = \frac{\beta_2}{\text{AvgPrice}_{it-1}}$$

In addition, after logarithmic transformation, our price measure follows a nearly normal distribution.

Estimates of model (2) are shown in column I of Table 2.[11] We applied Huber-White robust clustered standard errors to adjust for potential heteroskedasticity and repeated observations of the same cameras. We also examined the VIF of each covariate other than brand dummies and month dummies to detect multicollinearity and found all VIFs (except one resolution dummy, which is not the focus of this study) to be lower than 10. We also estimated fixed and random effects models. However, a Hausman test between fixed and random effects suggests that fixed effects are not necessary to remove bias, thus favoring random effects. This suggests that our camera attribute measures are doing an adequate job of controlling for product heterogeneity in our model. We therefore focus on pooled OLS results with Huber-White errors since this model provides a better fit, controls for potential correlation in error terms within cameras, and is not subject to undesirable parameter variation due to deviations of the actual error structure from the structure assumed by random effects.

The coefficients in Table 2 on the review components can be viewed as the relative weights consumers place on different quality attributes when determining an overall quality. Our results suggest that image quality has the highest weight, followed by construction, features, and ease of use. Using these estimates, we can construct a quality index for each camera using the Dpreview data; this quality index can then be used as a control in our analysis of CNet ratings. The best estimate for quality based on this analysis is

$$\text{DpQualityMeasure}_{it} = 0.32 \times \text{Construction}_{it} + 0.26 \times \text{Features}_{it} + 0.48 \times \text{ImageQuality}_{it} + 0.17 \times \text{EaseOfUse}_{it}$$

Coefficient of price in column I is negative and significant. Thus, hypothesis H1 is supported. This analysis also suggests that price plays a significant but not determinative role in the perception of value. If two cameras had the same observable attributes and quality, then a camera priced 20 percent lower would be perceived as being 0.08 higher in quality (a 40 percent difference in price would correspond to a 0.18 increase in perceived quality). These effects are substantial given that the standard deviation of value for money ratings of all cameras in our sample is only 0.34.

It should be noted that because our price measure is one period prior to the ratings, reverse causality should not be a concern. In fact, if the reverse relationship indeed plays a role in ratings, then it will make our result even stronger because the impact of ratings on prices should be positive, in contrast to the negative impact of prices on ratings that we found.

To corroborate that our model is capturing something unique about the value for money rating and not some general effect of any review, we repeat the analysis in column I for each of the other review components (see columns II through V, Table 2).[12] We find a general positive but not significant correlation between quality ratings and price consistent with economic predictions, our model assumptions, and consumers' expectations that high-quality products usually command higher prices. This analysis supports hypothesis H2 and also justifies the use of these four dimensions of reviews (construction, features, image quality, and ease of use) to measure consumers' perceptions of product quality.

Prices Effects on Unidimensional Ratings

In this section, we examine whether the ratings provided by single-dimension review sites are negatively influenced by price (hypothesis H3). In general, unconditional correlations between price and quality will generally be positive due to the

[11]The number of observations is less than the 862 listed in Table 1 because 62 cameras have reviews posted on Dpreviews.com during the first month after release for which we do not have observations for $AvgPrice_{it-1}$.

[12]In this corroboration analysis, some unexpected results are worth discussing even though they are not the focus of this study. For example, some camera attributes are positively correlated with features ratings, but not significantly or negatively correlated with image quality ratings, such as format, maximum shutter speed, and optical zoom. This seems to suggest that good add-ons or portability may affect consumers' views of camera features positively, but not necessarily their evaluation of image quality (e.g., brightness, color, and texture of the pictures). In addition, consumers may have higher expectations for cameras with these features, and therefore are more easily disappointed, which may offset the positive effects.

Table 2. Impact of Price on Different Dimensions of Dpreview Ratings

	Model (2)	Corroboration Analysis			
	I	II	III	IV	V
Dependent Variable	**$ValueForMoney_{it}$**	**$Construction_{it}$**	**$Features_{it}$**	**$ImageQuality_{it}$**	**$EaseofUse_{it}$**
$Construction_{it}$	0.32***		0.17**	0.41***	0.36***
	(0.09)		(0.09)	(0.13)	(0.08)
$Features_{it}$	0.26***	0.18*		0.61***	0.34***
	(0.09)	(0.10)		(0.10)	(0.10)
$ImageQuality_{it}$	0.48***	0.22***	0.32***		-0.02
	(0.06)	(0.06)	(0.08)		(0.07)
$EaseofUse_{it}$	0.17**	0.30***	0.27***	-0.03	
	(0.07)	(0.08)	(0.09)	(0.11)	
$Log(AvgPrice_{it-1})$	-0.36***	0.10	0.11	0.07	0.01
	(0.07)	(0.08)	(0.07)	(0.11)	(0.09)
$FormatDummy1_i$ (format = compact)	-0.09	-0.10	0.21***	-0.12	-0.10
	(0.08)	(0.07)	(0.06)	(0.11)	(0.07)
$FormatDummy2_i$ (format = ultra compact)	-0.13*	0.17*	0.20**	-0.23*	-0.15**
	(0.07)	(0.09)	(0.08)	(0.12)	(0.07)
$Log(1/MaxShutter_i)$	0.03	-0.00	0.16***	-0.06	-0.06
	(0.04)	(0.04)	(0.03)	(0.05)	(0.04)
$LCDSize_i$	0.05	-0.04	0.08	-0.10	-0.08
	(0.07)	(0.09)	(0.09)	(0.14)	(0.09)
$LCDResolution_i$	-0.00**	0.00	-0.00	-0.00	0.00*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
$DigitalZoom_i$	0.01	0.01	0.01	0.03	0.01
	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)
$OpticalZoom_i$	0.01*	-0.00	0.02***	-0.03**	-0.02**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
$StorageIncluded_i$	-0.00***	0.00	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
$BatteryChargerDummy_i$	-0.03	0.02	-0.22***	0.13	0.01
	(0.06)	(0.05)	(0.05)	(0.09)	(0.06)
$ManualFocusDummy_i$	0.12	0.05	0.09	0.08	-0.14**
	(0.07)	(0.07)	(0.09)	(0.11)	(0.07)
$ImageStabilizationDummy_i$	-0.15**	-0.11	0.14	-0.03	0.12
	(0.07)	(0.10)	(0.13)	(0.15)	(0.13)
$ResolutionDummy1_i$ (Resolution: 3 ~4 megapixels)	-0.06	-0.05	-0.11	0.11	0.34***
	(0.10)	(0.12)	(0.10)	(0.14)	(0.11)
$ResolutionDummy2_i$ (Resolution: 4 ~ 5 megapixels)	-0.15*	-0.07	0.01	0.06	0.22**
	(0.08)	(0.09)	(0.08)	(0.10)	(0.10)
$ResolutionDummy3_i$ (Resolution: 5 ~ 6 megapixels)	-0.10	-0.01	-0.08	0.01	0.30***
	(0.07)	(0.08)	(0.07)	(0.10)	(0.07)
$ResolutionDummy4_i$ (Resolution: 6 ~ 7 megapixels)	0.08	-0.09	-0.06	0.08	0.21**
	(0.09)	(0.08)	(0.08)	(0.08)	(0.08)
$ResolutionDummy5_i$ (Resolution: 7 ~ 8 megapixels)	-0.05	-0.06	-0.01	0.00	0.25***
	(0.07)	(0.08)	(0.09)	(0.09)	(0.08)
# of Observations	800	800	800	800	800
R-squared	0.835	0.708	0.800	0.727	0.634

The base case is FormatDummy3 = 1 (format = SLR) and ResolutionDummy6 = 1 (Resolution: 8 ~ 9 megapixels). Huber-White robust standard errors in parenthesis; *p < 0.1, **p < 0.05, ***p < 0.01; coefficients for brand dummies and month dummies are omitted from the table.

Table 3. Correlations between Ratings and Prices

Correlations	1	2	3	4	5
1. DPQualityMeasure	–				
2. Dpreview Value for Money Ratings	0.81	–			
3. CNet Editor Ratings	0.34	0.22	–		
4. CNet Consumer Ratings	0.35	0.35	0.17	–	
5. Price	0.45	0.16	0.36	0.15	–

simple fact that firms tend to charge higher prices for higher quality products. This pattern is shown in Table 3, in which we report simple correlations among prices, CNet ratings (a single-dimension review site), and Dpreview multidimensional ratings. We also include CNet editor ratings (unlike prices and other ratings, these do not vary over time, so we treat them as observed in the month they are posted).

Two observations are evident from Table 3. First, correlations between ratings and prices are all positive. High-quality products usually command high prices because it is costly to produce higher quality products and because consumers are willing to pay more for higher quality. Interestingly, this is true even for the value-for-money measure, suggesting that although price is a factor in value perceptions (as per column I in Table 2), the quality effects are dominant. Second, the correlation between CNet consumer ratings and prices is close to the correlation between Dpreview value ratings and prices, and both correlations are lower than that between prices and DPQualityMeasure (our constructed index of quality) or CNet editor ratings. Because we would expect the correlation between quality and price to be higher than the correlation between value and price, this observation suggests that CNet consumer ratings, similar to Dpreview value ratings, may be negatively affected by price, while CNet editor ratings are more likely to reflect quality instead of value.

We now examine the relationship between unidimensional reviews and prices, controlling for quality to test H3. We use two approaches to control for quality. First, we utilize our constructed quality measure, $\text{DPQualityMeasure}_{it}$.[13] This measure has three distinct advantages over other possible measures of quality: it is a consumer-based measure rather than based on firm claims; it varies across products and over time; and it is collected from a different web site than the one from which we collect the single-dimension ratings, thereby reducing the chance of common methods bias. Second, we construct an alternative measure of objective product quality from CNet editor ratings. Thus, we estimate variations of

$$\begin{aligned}\text{CNetConsumerRating}_{it} = {} & \beta_0 + \beta_{11}\text{DPQualityMeasure}_{it} \\ & + \beta_{12}\text{CNetEditorRating}_i + \beta_2\text{Log}(\text{AvgPrice}_{it-1}) + \alpha_1\text{FormatDum1}_i \\ & + \alpha_2\text{FormatDum2}_i + \alpha_3\text{Log}(1/\text{MaxShutter}_i) + \alpha_4\text{LCDSize}_i \\ & + \alpha_5\text{LCDResolution}_i + \alpha_6\text{DigitalZoom}_i + \alpha_7\text{OpticalZoom}_i \\ & + \alpha_8\text{StorageIncluded}_i + \alpha_9\text{BatteryChargerDummy}_i \\ & + \alpha_{10}\text{ManualFocusDummy}_i + \alpha_{11}\text{ImageStabilizationDum}_i \\ & + \lambda_1\text{ResolutionDummies}_i + \lambda_2\text{BrandDummies}_i \\ & + \lambda_3\text{MonthDummies}_t + \omega_{it}\end{aligned} \quad (3)$$

Regression results for model (3) are shown in Table 4. We focus on pooled OLS results with Huber-White robust clustered standard errors to account for repeated observations of the same camera over time. After we control for quality using both consumers' perceptions of quality (DPQuality Measure) and CNet editor ratings as well as other camera attributes, CNet consumer ratings are negatively affected by prices (column I in Table 4), supporting hypothesis H3. In particular, if two cameras had the same attributes and quality, one camera's average rating would be higher than the other one by 0.16 if its price was lower by 20 percent or 0.36 if its price was lower by 40 percent. These effects are substantial given that the standard deviation of CNet consumer ratings of all cameras in our sample is only 0.7. In addition, prior studies in the marketing literature suggest that consumers may not consider all products in their search, either due to limited awareness (Hauser and Wernerfelt 1990) or *a priori* elimination of certain products based on a limited set of features (Roberts and Lattin 1991). Thus, the variation of reviews for products a consumer might actually consider may be substantially less than 0.7, making the price effects correspondingly larger. For example, if a consumer considers an average rating of 4 and above to represent acceptable quality, then lowering the price by 10 percent may enable the product to enter the consumer's consideration set, which it could not enter previously under the higher price.

[13] To ensure that our averages are representative of an equilibrium measure of quality, we do not compute this measure until we have at least five reviews. We tried different cutoff values, and five provides the best fit. This restriction is relevant here but not in equation (2) because in the previous section, quality measures and value measures are rated by the same person.

Table 4. Role of Price in Unidimensional CNet Ratings

	Model (3)	Corroboration Analysis					
	I	II	III	IV	V	VI	VII
Dependent Variable	CNet Consumer Rating$_{it}$	CNet Consumer Rating$_{it}$	CNet Consumer Rating$_{it}$	CNet Consumer Rating$_{it}$	CNet Consumer Rating$_{it}$	CNet Editor Rating$_{it}$	CNet Consumer Rating$_{it}$
DPQualityMeasure$_{it}$	0.72***			0.68***		0.36*	0.45
	(0.20)			(0.20)		(0.20)	(0.28)
Construction$_{it}$					0.57**		
					(0.28)		
Features$_{it}$					-0.13		
					(0.35)		
ImageQuality$_{it}$					0.26		
					(0.29)		
EaseofUse$_{it}$					0.27		
					(0.25)		
CNetEditorRating$_{it}$	-0.07	0.08			-0.06		-0.07
	(0.12)	(0.12)			(0.12)		(0.17)
Log(AvgPrice$_{it-1}$)	-0.71**	-0.45	-0.42	-0.72**	-0.68**	0.30	-0.86**
	(0.29)	(0.32)	(0.32)	(0.30)	(0.30)	(0.37)	(0.38)
FormatDummy1$_i$	-0.31	-0.54**	-0.46**	-0.38*	-0.21	0.89**	-0.69**
(format = compact)	(0.21)	(0.21)	(0.21)	(0.20)	(0.24)	(0.36)	(0.34)
FormatDummy2$_i$	0.33	0.32	0.40	0.27	0.36	0.90**	0.31
(format = ultra compact)	(0.27)	(0.28)	(0.26)	(0.25)	(0.26)	(0.38)	(0.38)
Log(1/MaxShutter$_i$)	-0.09	-0.09	-0.05	-0.12	-0.06	0.42**	-0.13
	(0.15)	(0.16)	(0.14)	(0.13)	(0.15)	(0.20)	(0.23)
LCDSize$_i$	0.66**	0.55**	0.57**	0.63**	0.69**	0.59*	0.27
	(0.26)	(0.27)	(0.27)	(0.27)	(0.26)	(0.35)	(0.42)
LCDResolution$_i$	-0.00	-0.00	-0.00	-0.00	-0.00*	-0.00**	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
DigitalZoom$_i$	0.10	0.22***	0.21***	0.12*	0.10	-0.19	0.21*
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.12)	(0.11)
OpticalZoom$_i$	0.05*	0.03	0.03	0.05*	0.06**	0.04	0.02
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	(0.05)
StorageIncluded$_i$	0.00	0.00	0.00	0.00	0.00	0.01	-0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
BatteryChargerDummy$_i$	-0.08	-0.11	-0.13	-0.07	-0.15	-0.11	0.01
	(0.14)	(0.16)	(0.15)	(0.14)	(0.17)	(0.21)	(0.22)
ManualFocusDummy$_i$	0.99***	1.18***	1.21***	0.98***	1.03***	0.16	1.02***
	(0.20)	(0.23)	(0.22)	(0.20)	(0.20)	(0.27)	(0.30)
ImageStabilizationDummy$_i$	0.53*	0.61*	0.65**	0.50*	0.59**	0.38	0.78*
	(0.28)	(0.32)	(0.31)	(0.27)	(0.27)	(0.33)	(0.44)
ResolutionDummy1$_i$	-0.41	-0.12	-0.14	-0.38	-0.50*	-0.16	-0.27
(Resolution: 3 ~4 megapixels)	(0.29)	(0.34)	(0.34)	(0.30)	(0.29)	(0.50)	(0.40)
ResolutionDummy2$_i$	-0.46*	-0.31	-0.30	-0.46*	-0.50*	0.05	-0.55
(Resolution: 4 ~ 5 megapixels)	(0.27)	(0.31)	(0.30)	(0.28)	(0.28)	(0.46)	(0.42)
ResolutionDummy3$_i$	-0.54**	-0.39	-0.41	-0.52**	-0.63**	-0.09	-0.25
(Resolution: 5 ~ 6 megapixels)	(0.25)	(0.29)	(0.29)	(0.26)	(0.27)	(0.39)	(0.33)
ResolutionDummy4$_i$	0.37	0.57**	0.53**	0.42*	0.35	-0.40	0.40
(Resolution: 6 ~ 7 megapixels)	(0.23)	(0.24)	(0.23)	(0.23)	(0.23)	(0.47)	(0.33)
ResolutionDummy5$_i$	-0.20	-0.13	-0.14	-0.19	-0.24	0.17	-0.01
(Resolution: 7 ~ 8 megapixels)	(0.29)	(0.31)	(0.31)	(0.30)	(0.29)	(0.40)	(0.36)
# of Observations	727	727	727	727	727	69	69
R-squared	0.540	0.488	0.486	0.539	0.549	0.598	0.551

The base case is FormatDummy3 = 1 (format = SLR) and ResolutionDummy6 = 1 (Resolution: 8 ~ 9 megapixels). Huber-White robust standard errors in parenthesis; *p < 0.1, **p < 0.05, ***p < 0.01; coefficients for brand dummies and month dummies are omitted from the table.

We also estimated different variants of model (3). Our results are presented in columns II through V in Table 4. Consumers' perceptions of quality (DPQualityMeasure) appear to be a better control for quality than CNet editor rating. Inclusion of DPQualityMeasure can improve the fit of regression from ~49 percent to ~54 percent (see columns III and IV), while inclusion of the CNet editor rating does not substantially affect fit (see columns II and III). This is reasonable because when consumers write reviews, they express mostly their own perception of quality, rather than that of experts. This also explains why the coefficient of price in columns II and III is negative but not significant: camera features alone or with editor ratings are not sufficient controls for camera quality to test the price effects. Similar results on price effects hold if we incorporate the Dpreview quality components directly rather than using our quality index (see column V). This supplemental analysis further reveals that CNet consumer ratings appear to place higher weights on product construction.[14]

As a final corroboration of this analysis, we examine the relationship between CNet editor ratings and prices. The result is shown in column VI, Table 4. Unlike consumer ratings, CNet editor ratings are positively related to prices although not significant, suggesting that CNet editor ratings primarily reflect an editor's view of product quality rather than value of purchase.[15] Other than the price effects, some other camera attributes also seem to play different roles for consumer ratings versus editor ratings, suggesting that consumers and editors may have very different perspectives when reviewing products.

Empirical Tests of Model Predictions about Price Structure

Hypotheses H4 and H5 are tested with the following two regression models:

$$\begin{aligned}\text{MeanPrice}_i = {} & r_0 + r_1\text{DPQualityMeasure}_i + r_2\text{DPQualityMeasure}_i^2 \\ & + \beta_2\text{DateAfterRelease}_i + \alpha_1\text{FormatDum1}_i + \alpha_2\text{FormatDum2}_i \\ & + \alpha_3\text{Log}(1/\text{MaxShutter}_i) + \alpha_4\text{LCDSize}_i + \alpha_5\text{LCDResolution}_i \\ & + \alpha_6\text{DigitalZoom}_i + \alpha_7\text{OpticalZoom}_i + \alpha_8\text{StorageIncluded}_i \\ & + \alpha_9\text{BatteryChargerDummy}_i + \alpha_{10}\text{ManualFocusDummy}_i \\ & + \alpha_{11}\text{ImageStabilizationDummy}_i + \lambda_1\text{ResolutionDummies}_i \\ & + \lambda_2\ \text{BrandDummies}_i + \varepsilon_i\end{aligned} \quad (4)$$

$$\begin{aligned}\frac{\text{StdevPrice}_i}{\text{MeanPrice}_i} = {} & r_0 + r_2\text{DPQualityMeasure}_i + r_2\text{DPQualityMeasure}_i^2 \\ & + \beta_2\text{DateAfterRelease}_i + \alpha_1\text{FormatDum1}_i + \alpha_2\text{FormatDum2}_i \\ & + \alpha_3\text{Log}(1/\text{MaxShutter}_i) + \alpha_4\text{LCDSize}_i + \alpha_5\text{LCDResolution}_i \\ & + \alpha_6\text{DigitalZoom}_i + \alpha_7\text{OpticalZoom}_i + \alpha_8\text{StorageIncluded}_i \\ & + \alpha_9\text{BatteryChargerDummy}_i + \alpha_{10}\text{ManualFocusDummy}_i \\ & + \alpha_{11}\text{ImageStabilizationDummy}_i + \lambda_1\text{ResolutionDummies}_i \\ & + \lambda_2\text{BrandDummies}_i + \varepsilon_i\end{aligned} \quad (5)$$

MeanPrice_i represents the arithmetic mean of monthly prices for camera *i* from camera release date until the end of data collection. StdevPrice_i represents the standard deviation of monthly prices for camera *i* from camera release until the end of data collection and is normalized by MeanPrice_i to further control for product price heterogeneity unrelated to pricing strategy. We measure quality using our composite measure, $\text{DPQualityMeasure}_i$, calculated at the end of data collection, which from prior results appears to provide a good estimate of perceived consumer quality. This measure is entered linearly and with a square term to allow for the hypothesized U-shaped relationships. We control for how long a camera has been on the market in this analysis ($\text{DateAfterRelease}_i$), because standard deviation is expected to increase over time due to life cycle effects common to all products in the market.[16] Other control variables are constructed in the same way as in our previous analysis. Our estimating equations (4 and 5) can be interpreted directly as a structural estimate of the optimal pricing functions in the section on model setup (and model prediction (1) in the section on pricing, profitability, and rating) that relate price to quality and exogenous parameters, with controls for product heterogeneity included to account for pooling different cameras into a single analysis.

Regression results are presented in Table 5. After controlling for camera features, average price is a convex function of product quality as measured by DPQualityMeasure (see column I), while temporal variance of price over time is a concave function of quality (see column II). Thus hypotheses H4 and H5 are both supported. These findings are consistent with our model prediction that firms with middle-range quality pro-

[14]Although three of the four Dpreview quality components are not significant at 10% level individually, they are significant at 0.1% level together.

[15]Note that we have a considerably smaller sample in this analysis (69 observations) for two principal reasons: (1) there is no time variation in CNet editor ratings, and (2) further observations are lost when the editor ratings appear before there are sufficient (> 5) consumer ratings to construct our Dpreview quality rating. This sample reduction does not explain the change in sign on the price coefficient; when we run the analysis with CNet consumer ratings on a comparable sample, the coefficient in that analysis is still negative and significant (see Table 4, column VII).

[16]Including $\text{DateAfterRelease}_i$ greatly increases regression fit for model (5) from around 50% to around 80%, but does not affect model (4) much.

Table 5. Observed Price Patterns for Products with Different Quality

	Model (4)	Model (5)
	I	II
Dependent Variable	**Avg(P_i)**	**Stdev(P_i)/ Avg(P_i)**
$DPQualityMeasure_i$	-2,728.63**	0.74**
	(1,075.10)	(0.31)
$DPQualityMeasure_i^2$	279.72**	-0.07**
	(105.32)	(0.03)
$DateAfterRelease_i$	0.07	0.00***
	(0.06)	(0.00)
$FormatDummy1_i$ (format = compact)	-127.89**	0.01
	(62.00)	(0.03)
$FormatDummy2_i$ (format = ultra compact)	-155.59**	-0.00
	(66.08)	(0.03)
$Log(1/MaxShutter_i)$	65.40**	0.01
	(27.27)	(0.01)
$LCDSize_i$	138.10*	0.01
	(69.59)	(0.03)
$LCDResolution_i$	0.00***	-0.00
	(0.00)	(0.00)
$DigitalZoom_i$	-42.23**	-0.00
	(20.83)	(0.01)
$OpticalZoom_i$	7.36	-0.00
	(9.17)	(0.00)
$StorageIncluded_i$	-5.03***	0.00***
	(1.62)	(0.00)
$BatteryChargerDummy_i$	66.90	0.01
	(41.71)	(0.02)
$ManualFocusDummy_i$	-27.35	-0.00
	(49.30)	(0.03)
$ImageStabilizationDummy_i$	15.39	0.01
	(93.53)	(0.04)
$ResolutionDummy1_i$ (Resolution: 3 ~4 megapixels)	-221.33**	0.02
	(107.46)	(0.03)
$ResolutionDummy2_i$ (Resolution: 4 ~ 5 megapixels)	-178.38*	0.04
	(101.27)	(0.03)
$ResolutionDummy3_i$ (Resolution: 5 ~ 6 megapixels)	-89.40	-0.00
	(93.03)	(0.02)
$ResolutionDummy4_i$ (Resolution: 6 ~ 7 megapixels)	-46.58	-0.01
	(89.85)	(0.02)
$ResolutionDummy5_i$ (Resolution: 7 ~ 8 megapixels)	-5.97	-0.02
	(98.09)	(0.03)
Constant	6,348.59**	-1.88**
	(2,769.27)	(0.77)
# of Observations	88	88
R-squared	0.872	0.835

The base case is FormatDummy3 = 1 (format = SLR) and ResolutionDummy6 = 1 (Resolution: 8 ~ 9 megapixels). Huber-White robust standard errors in parenthesis; *p < 0.1, **p < 0.05, ***p < 0.01; coefficients for brand dummies and month dummies are omitted from the table.

ducts exhibit the highest incentive to undercut prices strategically to affect ratings. By validating our model predictions, this set of analyses is suggestive of supporting our argument that prices influence ratings.

Discussion

Prior literature has shown that consumer reviews play an important role in affecting consumers' purchase decisions (e.g., Chevalier and Mayzlin 2006; Dellarocas et al. 2004; Forman et al. 2008). When consumers make purchase decisions, both quality and price have an impact on their decision process. Quality is usually uncertain before purchase, but consumer reviews now provide a way of resolving, at least partially, this quality uncertainty. However, if these reviews are affected by the prices that reviewers paid for the product, which are usually unknown to the consumers who read the reviews, then the quality signal can be biased, and the decision that is made based on this biased information can be suboptimal. These observations may contribute to prior research on the trend of reviews over time (e.g., Li and Hitt 2008) by providing additional possible explanation that the dominant declining trend in average ratings over time may arise because early reviews are subject to price effects which may mislead later buyers, leading to disappointment and thus lowered ratings in later periods.

Different formats for review systems can influence the size of this bias. For systems that subdivide reviews into quality and value (or price) ratings, the bias in reviews due to price variation is likely to be minimal because consumers are able to observe quality directly from separate quality ratings. However, the most common review systems typically rely on a single rating which, as we show in the empirical analysis, can be substantially biased by price effects. Our results suggest that unidimensional ratings appear to be similar to a value measure in more complex rating systems. A comparison of our results on CNet and Dpreview suggest that the marginal impact of price is similar between the CNet overall rating and the Dpreview value rating after accounting for their different scales.

When consumers' perceptions can be influenced by price, firms have an additional strategic consideration in deciding how to position their products for sale. The theoretical results presented in the "Analytical Model" section suggest that for products that are of intermediate quality, firms may benefit by reducing their prices to boost their ratings in systems that cannot distinguish price from quality, although the existence of these price effects does not always benefit firms in that they may be worse off in a scenario in which this type of review bias is present. However, since firms do not have a choice as to whether these effects are present, only how they react to them, they are still better off pricing optimally to take price influences on reviews into account than if they ignore these effects. Consumers are also worse off as more consumers respond to imperfect information in their product choices.

There are two direct implications for our results. First, to the extent that single-dimension review systems are the norm, firms should account for price effects in their overall marketing strategy. This may be particularly important if other marketing dynamics are at play (such as word-of-mouth diffusion from lead adopters), because firms can potentially alter their product perceptions at critical times, which leads to substantially greater diffusion. A second implication is that review services, to the extent that they are attempting to promote greater consumer welfare, should consider expanding the review dimensions to reduce these price effects. If we calibrate our model parameters in the earlier section on consumer welfare using our empirical estimates[17] and utilize our quality measure DPQualityMeasure to construct the distribution of product quality for the cameras in our sample, we can estimate that the portion of cameras in our sample for which consumer surplus is negatively affected is between 11 percent (if n = 1, both periods have the same number of customers) and 100 percent (if n >9, later adopters make up most of the sample).

Although our results suggest that the price effects are substantial at least for the product category we consider, whether similar effects can be observed for other categories of products needs to be tested in future research. There are some other limitations of this work. First, because we can only observe the market-level average price data instead of the individual-level price paid by each reviewer, we are not able to associate each rating with the actual transaction price. As a result, we are forced to use aggregate measures for both ratings and prices (per-month cumulative average rating and price) instead of each single review. This limitation may, however, make our estimated coefficient of price more con-

[17]Estimates of model (3) in column I of Table 4 suggest $\partial \text{CNetConsumerRating}_{it} / \partial \text{AvgPrice}_{it-1} = -0.71 / \text{AvgPrice}_{it-1}$. According to Table 1, CNet consumer ratings range from 5 to 10 and average market prices range from $70 to $1,696. If we scale both ratings and prices to [0, 1] as assumed in our analytical model and utilize the mean of average market prices $435 (Table 1) as the base number on the right hand side of the equation, we can derive that the price effects parameter (b) is around 0.53. A sample of Comscore web visiting data (footnote 4) suggests that the portion of consumers who visit single-dimension review websites (value of parameter a) is above 90%.

servative, and indeed strengthen the argument that single-value ratings may largely reflect the consumer's perceived value from purchase instead of pure quality.

Second, we do not have sales data associated with reviews posted on Dpreview.com versus those posted on CNet.com, and hence we are not able to empirically compare the impact of reviews on sales across two different review formats (single-value and multiple-value ratings). Future study with finer-grained data may examine the indirect impact of price on sales through influencing consumer ratings and the extent to which the consumer's purchase behavior is affected by availability of separate quality ratings. In particular, it would be useful to know to what extent consumers are able to correct the bias in ratings caused by price.

Finally, in this paper, we use arrival time to separate early adopters and later followers and assume the total market size to be exogenous. Similar assumptions have also been used in previous studies of consumer reviews (e.g., Chen and Xie 2008). Whether reviews can affect consumers' redistribution across periods or affect the market size can be a fruitful direction for future study.

Acknowledgments

The authors would like to thank the Mack Center for Technology Innovation at Wharton for funding this research and thank the NPD Group and CIDRIS at University of Connecticut for data support. The authors would also like to thank the senior editor, Chris F. Kemerer, the associate editor, and the two anonymous reviewers for valuable comments and suggestions.

References

Anderson, E. W. 1998. "Customer Satisfaction and Word of Mouth," *Journal of Service* (1:1), pp. 5-17.

Bass, F. M. 1969. "A New Product Growth for Model Consumer Durables," *Management Science* (15:5), pp. 215-227.

Bolton, R. N., and Drew, J. H. 1991. "A Longitudinal Analysis of the Impact of Service Changes on Customer Attitudes," *Journal of Marketing* (55:1), pp. 1-10.

Bowman, D., and Narayandas, D. 2001. "Managing Customer-Initiated Contacts with Manufacturers: The Impact on Share of Category Requirements and Word-of-Mouth Behavior," *Journal of Marketing Research* (38:3), pp. 281-297.

Cadotte, E. R., Woodruff, R. B., and Jenkins, R. L. 1987. "Expectations and Norms in Models of Consumer Satisfaction," *Journal of Marketing Research* (24:3), pp. 305-314.

Chen, Y., and Xie, J. 2008. "Online Consumer Review: Word-of-Mouth as a New Element of Marketing Communication Mix," *Management Science* (54:3), pp. 477-490.

Chen. P., Dhanasobhon, S., and Smith, M. D. 2007. "All Reviews are Not Created Equal: The Disaggregate Impact of Reviews and Reviewers at Amazon.com," in *Proceedings of the 28th International Conference on Information Systems,* Montreal, Quebec, Canada, December 9-12/

Chevalier, J. A., and Mayzlin, D. 2006. "The Effect of Word of Mouth on Sales: Online Book Reviews," *Journal of Marketing Research* (43:3), pp. 345-354.

Churchill, G. A., and Surprenant, C. 1982. "An Investigation into the Determinants of Consumer Satisfaction," *Journal of Marketing Research* (19:4), pp. 491-504.

Clemons, E. K., Gao, G., and Hitt, L. M. 2006. "When Online Reviews Meet Hyperdifferentiation: A Study of the Craft Beer Industry," *Journal of Management Information Systems* (23:2), pp. 149-171.

ComScore. 2007. "Online Consumer-Generated Reviews Have Significant Impact on Offline Purchase Behavior," November 29 (available online at http://www.comscore.com/).

Dellarocas, C. 2006. "Strategic Manipulation of Internet Opinion Forums: Implications for Consumers and Firms," *Management Science* (52:10), pp.1577-1593.

Dellarocas, C, Awad, N. F., and Zhang, X. 2004. "Exploring the Value of Online Product Ratings in Revenue Forecasting: The Case of Motion Pictures," *Proceedings of the 25th International Conference on Information Systems,* R. Agarwal, L. Kirsch, and J. I. DeGross (eds.), Washington, DC, December 12-15, pp. 379-386.

DeGroot, M. H. 1970. *Probability and Statistics* (2nd ed.), Reading, MA: Addison-Wesley.

Deloitte. 2007. "New Deloitte Study Shows Inflection Point for Consumer Products Industry; Companies Must Learn to Compete in a More Transparent Age," Press Release, Deloitte Services LP, New York, October 1.

Dodds, W. B., Monroe, K. B., and Grewal, D. 1991. "Effects of Price, Brand, and Store Information on Buyers' Product Evaluations," *Journal of Marketing Research* (28:3), pp. 307-319.

DoubleClick. 2004. "DoubleClick's Touchpoints II: The Changing Purchase Process," March (available online at http://class.classmatandread.net/dct/dc_touchpoints_0403.pdf).

Duan, W., Gu, B., and Whinston, A. B. 2008. "The Dynamics of Online WOM and Product Sales–An Empirical Investigation of the Movie Industry," *Journal of Retailing* (84:2), pp. 233-242.

Forman, C., Ghose, A., and Wiesenfeld, B. 2008. "Examining the Relationship Between Reviews and Sales: The Role of Reviewer Identity Disclosure in Electronic Markets," *Information Systems Research* (19:3), pp. 291-313.

Foster, A. D., and Rosenzweig, M. R. 1995. "Learning by Doing and Learning from Others: Human Capital and Technical Change in Agriculture," *Journal of Political Economy* (103:6), pp. 1176-1209.

Gelman, A., and Hill, J. 2007. *Data Analysis Using Regression and Multilevel/Hierarchical Models*, Leiden, The Netherlands: Cambridge University Press.

Godes, D., and Mayzlin, D. 2004. "Using Online Conversations to Measure Word of Mouth Communication," *Marketing Science* (23:4), pp. 545-560.

Godes, D., Mayzlin D., Chen Y., Das S., Dellarocas C., Pfeiffer B., Libai, B., Sen, S., Shi, M., and Verlegh, P. 2005. "The Firm's Management of Social Interactions," *Marketing Letters* (16:3/4), pp. 415-428.

Grewal, D. 1995. "Product Quality Expectations: Towards an Understanding of Their Antecedents and Consequences," *Journal of Business and Psychology* (9:3), pp. 225-240.

Kirmani, A., and Rao, A. R. 2000. "No Pain, No Gain: A Critical Review of the Literature on Signaling Unobservable Product Quality," *Journal of Marketing* (64:2), pp. 66-79.

Hauser, J. R., and Wernerfelt, B. 1990. "An Evaluation Cost Model of Consideration Sets," *Journal of Consumer Research* (16:4), pp. 393-408.

Hotelling, H. 1929. "Stability in Competition," *The Economic Journal* (39), pp. 41-57.

Li, X., and Hitt, L. M. 2008. "Self Selection and Information Role of Online Product Reviews," *Information Systems Research* (19:4), pp. 456-474.

Li, X., Hitt, L. M., and Zhang, Z. J. 2010. "Product Reviews and Competition in Markets for Repeat Purchase Products," *Journal of Management Information Systems* (forthcoming).

Mahajan, V., and Muller, E. 1979. "Innovation Diffusion and New Product Growth Models in Marketing," *Journal of Marketing* (43:4), pp. 55-68.

Mayzlin, D. 2006. "Promotional Chat on the Internet," *Marketing Science* (25:2), pp.155-163.

McGregor, J., Jespersen, F. F., Tucker, M., and Foust, D. 2007. "Customer Service Champs," *Business Week*, March 5 (available at http://www.businessweek.com/magazine/content/07_10/b4024001.htm).

Mitra, A. 1995. "Price Cue Utilization in Product Evaluations: The Moderating Role of Motivation and Attribute Information," *Journal of Business Research* (33), pp. 187-195.

Olson, J. C. 1997. "Price as an Informational Cue: Effects on Product Evaluations," in *Consumer and Industrial Buying Behaviour*, A. G. Woodside, J. N. Sheth, and P. D. Bennett (eds.), Amsterdam: North-Holland Publishing Company, pp. 267-86.

Piller, C. 1999. "Everyone Is a Critic in Cyberspace," *Los Angeles Times*, December 3 (available at http://articles.latimes.com/1999/dec/03/news/mn-40120).

Rao, A. R., and Monroe, K. B. 1988. "The Moderating Effect of Prior Knowledge on Cue Utilization in Product Evaluations," *The Journal of Consumer Research* (15:2), pp. 253-264.

Rao, A. R., and Monroe, K. B. 1989. "The Effect of Price, Brand Name, and Store Name on Buyers' Perceptions of Product Quality: An Integrative Review," *Journal of Marketing Research* (25:3), pp. 351-357.

Reingen, P. H., Foster, B. L., Brown, J. J., and Seidman, S. B. 1984. "Brand Congruence in Interpersonal Relations: a Social Network Analysis," *Journal of Consumer Research* (11:3), pp. 771-783

Richins, M. L. 1983. "Negative Word-of-Mouth by Dissatisfied Consumers: A Pilot Study," *Journal of Marketing* (47:1), pp. 68-78.

Roberts, J. H., and Lattin, J. M. 1991. "Development and Testing of a Model of Consideration Set Composition," *Journal of Marketing Research* (28:4), pp. 429-441.

Rogers, E. M. 1962. *Diffusion of Innovations*, New York: Free Press.

Rust, R. T., Inman, J. J., Jia, J., and Zahorik, A. 1999. "What You Don't Know about Customer-Perceived Quality: The Role of Customer Expectation Distributions," *Marketing Science* (18:1), pp. 77-92.

Schmalensee, R. 1982. "Product Differentiation Advantages of Pioneering Brands," *The American Economic Review* (72:3), pp. 349-365.

Shapiro, C. 1983a. "Optimal Pricing of Experience Goods," *The Bell Journal of Economics* (14:2), pp. 497-507.

Shapiro, C. 1983b. "Premiums for High Quality Products as Returns to Reputations," *The Quarterly Journal of Economics* (98:4), pp. 659-680.

Spreng R. A, MacKenzie, S. B., and Olshavsky, R. W. 1996. "A Reexamination of the Determinants of Consumer Satisfaction," *Journal of Marketing* (60:3), pp. 15-32.

Tsai, J. 2007. "Customer Satisfaction's Durability in Question," *destinationCRM.com*, November 20 (available online at http://www.destinationcrm.com/Articles/CRM-News/Daily-News/Customer-Satisfactions-Durability-in-Question-43512.aspx).

Varki, S., and Colgate, M. 2001. "Role of Price Perceptions in an Integrated Model of Behavioural Intentions," *Journal of Service Research* (3:3), pp. 232-240.

Villas-Boas, M. J. 2004. "Consumer Learning, Brand Loyalty, and Competition," *Marketing Science* (23:1), pp. 134-145.

Voss, G. B., Parasuraman, A., and Grewal, D. 1998. "The Roles of Price Performance, and Expectations in Determining Satisfaction in Service Exchanges," *Journal of Marketing* (62:4), pp. 46-61.

Zeithaml, V. A. 1988. "Consumer Perceptions of Price, Quality and Value: A Means-End Model and Synthesis of Evidence," *Journal of Marketing* (52:3), pp. 2-22.

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MIS Quarterly

RESEARCH ARTICLE

PRICE EFFECTS IN ONLINE PRODUCT REVIEWS: AN ANALYTICAL MODEL AND EMPIRICAL ANALYSIS

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Appendix A

Derivation of Optimal Price Functions for the Monopoly Setting

We apply backward induction to derive optimal price functions. In the second period, given first period price p_1, the firm selects second period price p_2 ($p_2 < Max\{q, R\}$) to maximize its second period profit:

$$\pi_2 = n\left((1-a)p_2 \text{Max}\left\{0, \frac{q-p_2}{t}\right\} + ap_2 \text{Max}\left\{0, \frac{\text{Max}\left\{0, \text{Min}\left\{1, q-b\left(p_1-\frac{q}{2}\right)\right\}\right\}-p_2}{t}\right\}\right)$$

The profit function can be reduced to four possibilities based on the value of p_1:

$$\pi_2 = \begin{cases} n\left((1-a)p_2 \text{Max}\left\{0, \frac{q-p_2}{t}\right\} + ap_2 \frac{1-p_2}{t}\right) & if\ 0 < p_1 < \frac{(2+b)q-2}{2b} \\ n\left((1-a)p_2 \text{Max}\left\{0, \frac{q-p_2}{t}\right\} + ap_2 \frac{q-b\left(p_1-\frac{q}{2}\right)-p_2}{t}\right) & if\ \frac{(2+b)q-2}{2b} \le p_1 < \frac{q}{2} \\ n\left((1-a)p_2 \frac{q-p_2}{t} + ap_2 \text{Max}\left\{0, \frac{q-b\left(p_1-\frac{q}{2}\right)-p_2}{t}\right\}\right) & if\ \frac{q}{2} \le p_1 \le \frac{(2+b)q}{2b} \\ n\left((1-a)p_2 \frac{q-p_2}{t}\right) & if\ \frac{(2+b)q}{2b} < p_1 < q^e \end{cases}$$

By maximizing profit in each of the four cases, we can derive the optimal second period price p_2 as a function of first period price p_1:

$$p_2^*(p_1)=\begin{cases}\dfrac{a+(1-a)q}{2} & if\ 0<p_1<\dfrac{(2+b)q-2}{2b}\\ \dfrac{q}{2}+\dfrac{ab(q-2p_1^*)}{4} & if\ \dfrac{(2+b)q-2}{2b}<p_1<\dfrac{(2-2\sqrt{1-a}+ab)q}{2ab}\\ \dfrac{q}{2} & if\ \dfrac{(2-2\sqrt{1-a}+ab)q}{2ab}<p_1<q^e\end{cases}$$

The corresponding second period profit as a function of p_1 is

$$\pi_2^*(p_1)=\begin{cases}\dfrac{n(a+(1-a)q)^2}{4t} & if\ 0<p_1<\dfrac{(2+b)q-2}{2b}\\ \dfrac{n\left(2q+ab(q-2p_1)\right)^2}{16t} & if\ \dfrac{(2+b)q-2}{2b}<p_1<\dfrac{(2-2\sqrt{1-a}+ab)q}{2ab}\\ \dfrac{n(1-a)q^2}{4t} & if\ \dfrac{(2-2\sqrt{1-a}+ab)q}{2ab}<p_1<q^e\end{cases}$$

Back in the first period, given $\pi_2^*(p_1)$, the firm selects a first period price p_1 ($p_1 < q^e$) to maximize its total profit in both periods: $\frac{p_1(q^e-p_1)}{t}+\pi_2^*(p_1)$. By comparing optimal profit in different ranges of p_1, we can derive the optimal first period price for different values of q:

$$p_1^*=\begin{cases}\dfrac{q^e}{2} & \text{if } 0<q\le\bar{Q}_1 \text{ or } \dfrac{2+bq^e}{2+b}\le q<1\\ \text{Max}\left\{0,\dfrac{(2+b)q-2}{2b}\right\} & \text{if Max}\left\{\bar{Q}_1,\text{Min}\left\{\bar{Q}_2,\dfrac{4q^e}{2abn+a^2b^2n}\right\}\right\}\le q<\dfrac{2+bq^e}{2+b}\\ \dfrac{q^e}{2}-\dfrac{ab((2+ab)q-abq^e)n}{2(4-a^2b^2n)} & \text{if } \bar{Q}_1<q<\text{Max}\left\{\bar{Q}_1,\text{Min}\left\{\bar{Q}_2,\dfrac{4q^e}{2abn+a^2b^2n}\right\}\right\}\end{cases}$$

Combining p_1^* and $p_2^*(p_1)$, we can derive that $p_2^*=\begin{cases}\dfrac{a+(1-a)q}{2} & \text{if Max}\left\{\bar{Q}_2,\dfrac{2}{2+b}\right\}<q<1\\ \dfrac{q}{2}+\dfrac{ab(q-2p_1^*)}{4} & \text{if } \bar{Q}_1<q<\text{Max}\left\{\bar{Q}_2,\dfrac{2}{2+b}\right\}\\ \dfrac{q}{2} & \text{if } 0<q<\bar{Q}_1\end{cases}$.

Appendix B

Derivation of Optimal Price Functions for the Duopoly Setting

We first utilize the case of $q_1=\frac{1}{2}$ to explain in detail how to derive equilibrium prices and then follow the same procedure to solve equilibria for the other two cases ($q_1 = 1$ and $q_1 = 0$).

We apply backward induction to derive optimal price functions. In the second period, given the first period prices, p_{11} and p_{21} ($p_{j1} < q_j^e = \frac{1}{2}$), the ratings of the two products are $R_1 = Min\left\{1, Max\left\{0, \frac{3-4p_{11}}{4}\right\}\right\}$ and $R_2 = Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}$. Given $a = 1$, $b = 1$, $n = 3$, $t = \frac{1}{6}$, and $q_1^e = q_2^e = \frac{1}{2}$, if all second-period consumers purchase from one of the two firms, the second period profits are $\pi_{12} = 3\left(p_{12}\,Min\left\{1, Max\left\{0, 3\left(R_1 - R_2 - p_{12} + p_{22} + \frac{1}{6}\right)\right\}\right\}\right)$ and $\pi_{22} = 3\left(p_{22}\,Min\left\{1, Max\left\{0, 3\left(R_2 - R_1 - p_{22} + p_{12} + \frac{1}{6}\right)\right\}\right\}\right)$. If some second-period consumers expect negative utility from both firms and do not buy from either firm, the profit functions are $\pi_{12} = 3\left(p_{12}\,Min\left\{1, Max\left\{0, 6\left(R_1 - p_{12}\right)\right\}\right\}\right)$ and $\pi_{22} = 3\left(p_{22}\,Min\left\{1, Max\left\{0, 6\left(R_2 - p_{22}\right)\right\}\right\}\right)$. Then back in the first period, firms select p_{11} and p_{21} to maximize their total profits in both periods: $\pi_1 = 3p_{11}\,Min\left\{1, Max\left\{0, -p_{11} + p_{21} + \frac{1}{6}\right\}\right\} + \pi_{12}^*$ and

$\pi_2 = 3p_{21}\,Min\left\{1, Max\left\{0, -p_{21} + p_{11} + \frac{1}{6}\right\}\right\} + \pi_{22}^*$. It can be proved that in this scenario all of the second-period consumers will purchase one of the two products in equilibrium. Thus, firms' second period profit functions are:

$$\pi_{12} = 3\left(p_{12}\,Min\left\{1, 3\left(\frac{3-4p_{11}}{4} - Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\} - p_{12} + p_{22} + \frac{1}{6}\right)\right\}\right),$$

$$\pi_{22} = 3\left(p_{22}\,Max\left\{0, 3\left(Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\} - \frac{3-4p_{11}}{2} - p_{22} + p_{12} + \frac{1}{6}\right)\right\}\right).$$

We can then derive the optimal second period prices as functions of the first period prices:

$$p_{12}^*(p_{11}, p_{21}) = \begin{cases} 0 & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} > \frac{5}{4} \\ \frac{5-4\left(p_{11}+Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)}{3} & if\ \frac{1}{4} < p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{5}{4}, \\ \frac{7-12\left(p_{11}+Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)}{12} & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{1}{4} \end{cases}$$

$$p_{22}^*(p_{11}, p_{21}) = \begin{cases} \frac{-11-12\left(p_{11}+Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)}{12} & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} > \frac{5}{4} \\ \frac{-1+4\left(p_{11}+Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)}{12} & if\ \frac{1}{4} < p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{5}{4}. \\ 0 & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{1}{4} \end{cases}$$

The corresponding second period profits as functions of the first period prices thus are:

$$\pi_{12}^*(p_{11}, p_{21}) = \begin{cases} 0 & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} > \frac{5}{4} \\ \frac{\left(\frac{5-4p_{11}}{4} - Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)^2}{3} & if\ \frac{1}{4} < p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{5}{4}, \\ \frac{7-12\left(p_{11}+Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)}{4} & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{1}{4} \end{cases}$$

$$\pi_{22}^*(p_{11}, p_{21}) = \begin{cases} \frac{-11-12\left(p_{11}+Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)}{4} & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} > \frac{5}{4} \\ \frac{\left(\frac{1-4p_{11}}{4} - Min\left\{1, Max\left\{0, \frac{3q_2-2p_{21}}{2}\right\}\right\}\right)^2}{3} & if\ \frac{1}{4} < p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{5}{4}. \\ 0 & if\ p_{11} + Min\left\{1, Max\left\{0, \frac{3q_2}{2} - p_{21}\right\}\right\} < \frac{1}{4} \end{cases}$$

Then back in the first period, firms select the first period prices to maximize their total profits in both periods:

$$\pi_1 = 3p_{11}\,Min\left\{1, Max\left\{0, -p_{11} + p_{21} + \frac{1}{6}\right\}\right\} + \pi_{12}^*\left(p_{11}, p_{21}\right),$$

$$\pi_2 = 3p_{21}\,Min\left\{1, Max\left\{0, -p_{21} + p_{11} + \frac{1}{6}\right\}\right\} + \pi_{22}^*\left(p_{11}, p_{21}\right).$$

By comparing profits in different ranges of p_{11} and p_{21}, we can derive the optimal first period prices for different values of q_2:

$$p_{11}^* = \begin{cases} \frac{1}{10} & if\ \frac{34}{45} < q_2 < 1 \\ \frac{3(3q_2-2)}{8} & if\ \frac{2}{3} < q_2 < \frac{34}{45} \\ 0 & if\ 0 < q_2 < \frac{2}{3} \end{cases}, p_{21}^* = \begin{cases} \frac{2}{15} & if\ \frac{34}{45} < q_2 < 1 \\ \frac{3q_2-2}{2} & if\ \frac{2}{3} < q_2 < \frac{34}{45} \\ 0 & if\ \frac{1}{3} < q_2 < \frac{2}{3} \\ \frac{-3q_2+1}{4} & if\ \frac{2}{9} < q_2 < \frac{1}{3} \\ \frac{1}{12} & if\ 0 < q_2 < \frac{2}{9} \end{cases}.$$

Combining p_{11}^*, p_{21}^*, $p_{12}^*(p_{11}, p_{21})$, and $p_{22}^*(p_{11}, p_{21})$, we can derive that:

$$p_{12}^* = \begin{cases} \frac{1}{20} & if\ \frac{34}{45} < q_2 < 1 \\ \frac{8-9q_2}{24} & if\ \frac{2}{3} < q_2 < \frac{34}{45} \\ \frac{5-6q_2}{12} & if\ \frac{1}{3} < q_2 < \frac{2}{3} \\ \frac{1}{2} - \frac{3q_2}{4} & if\ \frac{2}{9} < q_2 < \frac{1}{3} \\ \frac{2}{3} - \frac{3q_2}{2} & if\ \frac{1}{18} < q_2 < \frac{2}{9} \\ \frac{7}{12} & if\ 0 < q_2 < \frac{1}{18} \end{cases},\ p_{22}^* = \begin{cases} \frac{17}{60} & if\ \frac{34}{45} < q_2 < 1 \\ \frac{3q_2}{8} & if\ \frac{2}{3} < q_2 < \frac{34}{45} \\ \frac{6q_2-1}{12} & if\ \frac{1}{3} < q_2 < \frac{2}{3} \\ -\frac{1}{6} + \frac{3q_2}{4} & if\ \frac{2}{9} < q_2 < \frac{1}{3} \\ 0 & if\ 0 < q_2 < \frac{2}{9} \end{cases}.$$

Following similar procedure, we can derive the optimal price functions for $q_1 = 1$:

$$p_{11}^* = \begin{cases} \frac{1}{6} & if\ \frac{7}{9} < q_2 < 1 \\ \frac{9q_2-5}{12} & if\ \frac{2}{3} < q_2 < \frac{7}{9} \\ \frac{1}{12} & if\ \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{13-18q_2}{30} & if\ \frac{4}{9} < q_2 < \frac{7}{12} \\ \frac{1}{6} & if\ 0 < q_2 < \frac{4}{9} \end{cases},\ p_{21}^* = \begin{cases} \frac{1}{6} & if\ \frac{7}{9} < q_2 < 1 \\ \frac{3q_2-2}{2} & if\ \frac{2}{3} < q_2 < \frac{7}{9} \\ 0 & if\ \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{7-12q_2}{10} & if\ \frac{4}{9} < q_2 < \frac{7}{12} \\ \frac{1}{6} & if\ 0 < q_2 < \frac{4}{9} \end{cases}.$$

$$p_{12}^* = \begin{cases} \frac{1}{6} & if\ \frac{2}{3} < q_2 < 1 \\ \frac{1-q_2}{2} & if\ \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{22-27q_2}{30} & if\ \frac{4}{9} < q_2 < \frac{7}{12} \\ 1 - \frac{3q_2}{2} & if\ 0 < q_2 < \frac{4}{9} \end{cases},\ p_{22}^* = \begin{cases} \frac{1}{6} & if\ \frac{2}{3} < q_2 < 1 \\ \frac{3q_2-1}{6} & if\ \frac{7}{12} < q_2 < \frac{2}{3} \\ \frac{9q_2-4}{10} & if\ \frac{4}{9} < q_2 < \frac{7}{12} \\ 0 & if\ 0 < q_2 < \frac{4}{9} \end{cases}.$$

Similarly, the optimal price functions for $q_1 = 0$ are

$$p_{11}^* = \begin{cases} \frac{1}{6} & if\ \frac{7}{9} < q_2 < 1 \\ \frac{9q_2-5}{12} & if\ \frac{2}{3} < q_2 < \frac{7}{9} \\ \frac{1}{12} & if\ \frac{2}{9\sqrt{6}} < q_2 < \frac{2}{3} \\ \frac{1}{6} & if\ 0 < q_2 < \frac{2}{9\sqrt{6}} \end{cases},\ p_{21}^* = \begin{cases} \frac{1}{6} & if\ \frac{7}{9} < q_2 < 1 \\ \frac{3q_2-2}{2} & if\ \frac{2}{3} < q_2 < \frac{7}{9} \\ 0 & if\ \frac{2}{9\sqrt{6}} < q_2 < \frac{2}{3} \\ \frac{1}{6} & if\ 0 < q_2 < \frac{2}{9\sqrt{6}} \end{cases}.$$

$$p_{12}^* = 0,\ p_{22}^* = \begin{cases} \frac{5}{6} & if\ \frac{2}{3} < q_2 < 1 \\ \frac{3q_2}{2} - \frac{1}{6} & if\ \frac{2}{9} < q_2 < \frac{2}{3} \\ \frac{3q_2}{4} & if\ \frac{2}{9\sqrt{6}} < q_2 < \frac{2}{9} \\ 0 & if\ 0 < q_2 < \frac{2}{9\sqrt{6}} \end{cases}.$$

In the benchmark scenario, the firms select p_{11}, p_{21}, p_{12}, and p_{22} to maximize their total profits:

$$\pi_1 = 3p_{11}\,Min\left\{1, Max\left\{0, -p_{11} + p_{21} + \tfrac{1}{6}\right\}\right\} + 3\left(p_{12}\,Min\left\{1, Max\left\{0, 3\left(q_1 - q_2 - p_{12} + p_{22} + \tfrac{1}{6}\right)\right\}\right\}\right),$$

$$\pi_2 = 3p_{21}\,Min\left\{1, Max\left\{0, -p_{21} + p_{11} + \tfrac{1}{6}\right\}\right\} + 3\left(p_{22}\,Min\left\{1, Max\left\{0, 3\left(q_2 - q_1 = p_{22} + p_{12} + \tfrac{1}{6}\right)\right\}\right\}\right).$$

It can be shown that the optimal first period prices are both $\frac{1}{6}$, the second period prices are:

(1) If $q_1 = 1, p_{21}^* = \begin{cases} \frac{1}{6} + \frac{1-q_2}{3} & if\ \frac{1}{2} < q_2 < 1 \\ \frac{5}{6} - q_2 & if\ 0 < q_2 < \frac{1}{2} \end{cases}, p_{22}^* = \begin{cases} \frac{1}{6} - \frac{1-q_2}{3} & if\ \frac{1}{2} < q_2 < 1 \\ 0 & if\ 0 < q_2 < \frac{1}{2} \end{cases}$.

(2) If $q_1 = \frac{1}{2}, p_{21}^* = \frac{1}{6} + \frac{\frac{1}{2} - q_2}{3}, p_{22}^* = \frac{1}{6} - \frac{\frac{1}{2} - q_2}{3}$.

(3) If $q_1 = 0, p_{21}^* = 0, p_{22}^* = \begin{cases} \frac{q_2}{2} & if\ 0 < q_2 < \frac{1}{3} \\ q_2 - \frac{1}{6} & if\ \frac{1}{3} < q_2 < 1 \end{cases}$.